

A Computational Model of Allophony as Dispersion over Phonetic Sequences



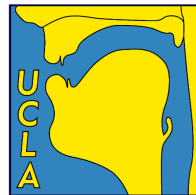
John McGahay

University of California, Los Angeles

Presented at LSA 2026

January 11, 2026

New Orleans, LA



Dispersion Theory

Dispersion Theory (Liljencrants and Lindblom, 1972): world's most common sound systems exhibit maximal perceptual contrast

Dispersion Theory

Dispersion Theory (Liljencrants and Lindblom, 1972): world's most common sound systems exhibit maximal perceptual contrast

- e.g. 'corner' vowels /iau/ found in most systems minimize confusion

Dispersion Theory

Dispersion Theory (Liljencrants and Lindblom, 1972): world's most common sound systems exhibit maximal perceptual contrast

- e.g. 'corner' vowels /iau/ found in most systems minimize confusion
- Systems optimizing psychoacoustic distance between vowels match typology
 - e.g. /iau/, /ieaou/, etc.

Dispersion Theory

Dispersion Theory (Liljencrants and Lindblom, 1972): world's most common sound systems exhibit maximal perceptual contrast

- e.g. 'corner' vowels /iau/ found in most systems minimize confusion
- Systems optimizing psychoacoustic distance between vowels match typology
 - e.g. /iau/, /ieaou/, etc.
- Many subsequent versions
 - e.g. Lindblom (1986), Schwartz et al. (1997b), Flemming (1995, 2004)

Dispersion Theory

Dispersion Theory (Liljencrants and Lindblom, 1972): world's most common sound systems exhibit maximal perceptual contrast

- e.g. 'corner' vowels /iau/ found in most systems minimize confusion
- Systems optimizing psychoacoustic distance between vowels match typology
 - e.g. /iau/, /ieaou/, etc.
- Many subsequent versions
 - e.g. Lindblom (1986), Schwartz et al. (1997b), Flemming (1995, 2004)
- Generally restricted to predicting inventories of individual segments, especially vowels

Sensitivity of Perception to Larger Sequences

Perception of individual speech sounds is sensitive to properties of larger sequences:

Sensitivity of Perception to Larger Sequences

Perception of individual speech sounds is sensitive to properties of larger sequences:

- wordhood (Ganong, 1980)
 - Ambiguous stop perceived as /g/ in /_ift/ but /k/ in /_is/ since *gift* and *kiss* are words of English but not *kift* and *giss*

Sensitivity of Perception to Larger Sequences

Perception of individual speech sounds is sensitive to properties of larger sequences:

- wordhood (Ganong, 1980)
 - Ambiguous stop perceived as /g/ in /_ɪft/ but /k/ in /_ɪs/ since *gift* and *kiss* are words of English but not *kift* and *giss*
- phonotactic legality (Massaro and Cohen, 1983)
 - Ambiguous liquid perceived as /l/ after /s/ but /ɹ/ after /t/ since /sl/ and /tɹ/ are legal English onsets but not /*sɹ/ and /*tl/

Sensitivity of Perception to Larger Sequences

Perception of individual speech sounds is sensitive to properties of larger sequences:

- wordhood (Ganong, 1980)
 - Ambiguous stop perceived as /g/ in /_ɪft/ but /k/ in /_ɪs/ since *gift* and *kiss* are words of English but not *kift* and *giss*
- phonotactic legality (Massaro and Cohen, 1983)
 - Ambiguous liquid perceived as /l/ after /s/ but /ɹ/ after /t/ since /sl/ and /tɹ/ are legal English onsets but not /*sɹ/ and /*tl/
- biphone frequency (Steffman and Sundara, 2024)
 - Exposure to biphones at different frequencies induces categorization bias towards more frequent biphone

Bayesian Models of Speech Perception

Such sequence effects look Bayesian

Bayesian Models of Speech Perception

Such sequence effects look Bayesian

- When the bottom-up acoustic signal is ambiguous, listeners rely on prior bias from top-down knowledge of the lexicon and phonotactics

Bayesian Models of Speech Perception

Such sequence effects look Bayesian

- When the bottom-up acoustic signal is ambiguous, listeners rely on prior bias from top-down knowledge of the lexicon and phonotactics

Human speech perception is well approximated by Bayesian models of optimal listeners

- cf. Clayards et al. (2008); Norris and McQueen (2008); Feldman et al. (2009); Kleinschmidt (2019)

Bayesian Models of Speech Perception

Such sequence effects look Bayesian

- When the bottom-up acoustic signal is ambiguous, listeners rely on prior bias from top-down knowledge of the lexicon and phonotactics

Human speech perception is well approximated by Bayesian models of optimal listeners

- cf. Clayards et al. (2008); Norris and McQueen (2008); Feldman et al. (2009); Kleinschmidt (2019)

Suggests potential for computational modeling of perceptual pressures due to sequence effects

Dispersion over Phonetic Sequences

Proposal: perceptually optimal sound systems optimize not only segments but also sequences

Dispersion over Phonetic Sequences

- Proposal: perceptually optimal sound systems optimize not only segments but also sequences
- Not new; Steriade (1993, 1997) attributes phonotactic patterns to perceptual pressures

Dispersion over Phonetic Sequences

Proposal: perceptually optimal sound systems optimize not only segments but also sequences

- Not new; Steriade (1993, 1997) attributes phonotactic patterns to perceptual pressures
 - Perceptual pressures encoded in OT constraints, not computational model of perception

Dispersion over Phonetic Sequences

Proposal: perceptually optimal sound systems optimize not only segments but also sequences

- Not new; Steriade (1993, 1997) attributes phonotactic patterns to perceptual pressures
 - Perceptual pressures encoded in OT constraints, not computational model of perception

Iterated Confusion Minimization (McGahay, 2025)

- Algorithm simulating interaction of optimal listeners and speakers over time

Optimal Listener Behavior

Optimal Listener Behavior

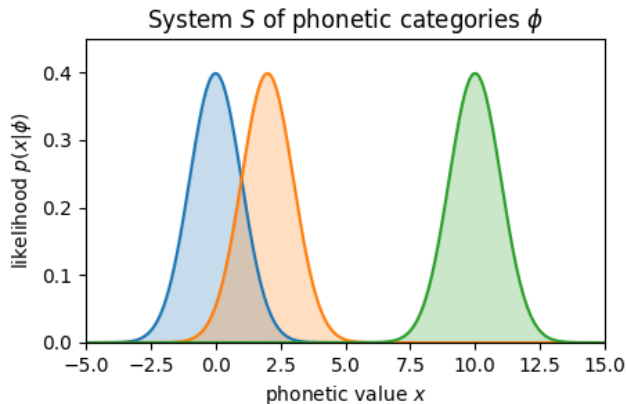
Simplifying assumption: phonetic categories have equal prior probabilities

Optimal Listener Behavior

Simplifying assumption: phonetic categories have equal prior probabilities

⇒ Optimal Bayesian listener follows **maximum-likelihood behavior**

Optimal Listener Behavior

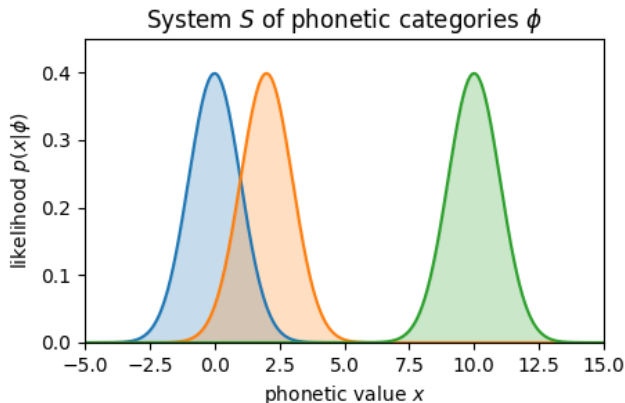


Optimal Listener Behavior

Tokens classified as category with highest likelihood at phonetic value

maximum-likelihood behavior

$$C(x) = \operatorname{argmax}_{\phi \in S} p(x|\phi)$$

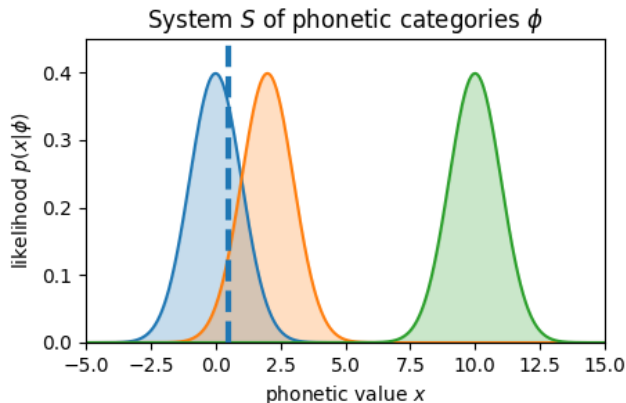


Optimal Listener Behavior

Tokens classified as category with highest likelihood at phonetic value

maximum-likelihood behavior

$$C(x) = \operatorname{argmax}_{\phi \in S} p(x|\phi)$$

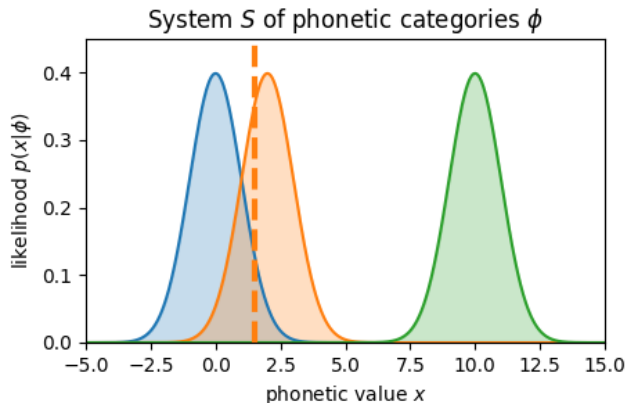


Optimal Listener Behavior

Tokens classified as category with highest likelihood at phonetic value

maximum-likelihood behavior

$$C(x) = \operatorname{argmax}_{\phi \in S} p(x|\phi)$$



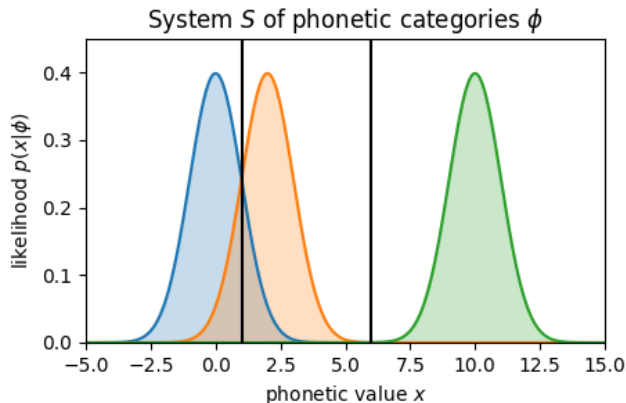
Optimal Listener Behavior

Tokens classified as category with
highest likelihood at phonetic value

⇒ categorical perceptual boundaries

maximum-likelihood behavior

$$C(x) = \operatorname{argmax}_{\phi \in S} p(x|\phi)$$



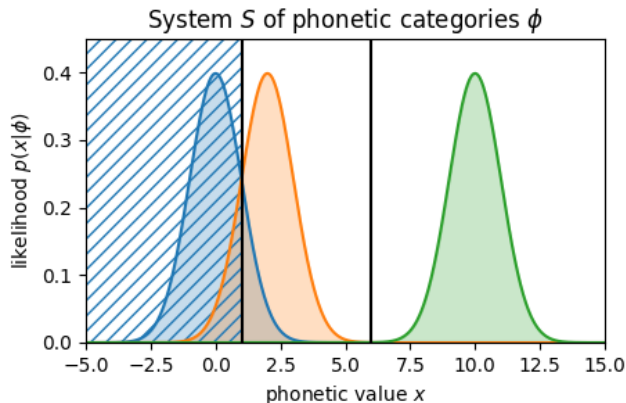
Optimal Listener Behavior

Tokens classified as category with
highest likelihood at phonetic value

⇒ categorical perceptual boundaries

maximum-likelihood behavior

$$C(x) = \operatorname{argmax}_{\phi \in S} p(x|\phi)$$



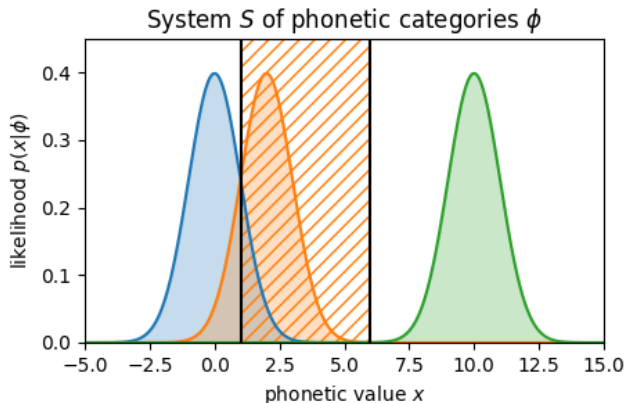
Optimal Listener Behavior

Tokens classified as category with highest likelihood at phonetic value

⇒ categorical perceptual boundaries

maximum-likelihood behavior

$$C(x) = \operatorname{argmax}_{\phi \in S} p(x|\phi)$$



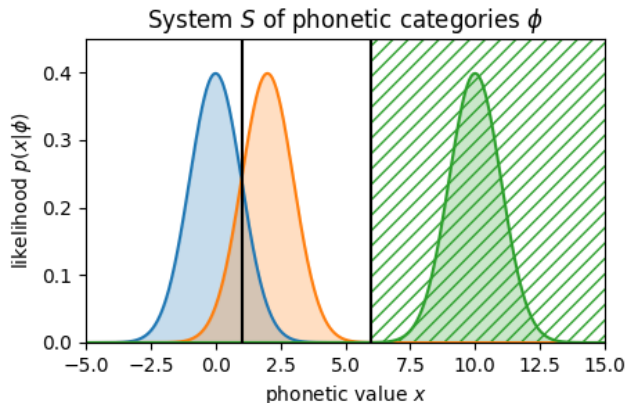
Optimal Listener Behavior

Tokens classified as category with highest likelihood at phonetic value

⇒ categorical perceptual boundaries

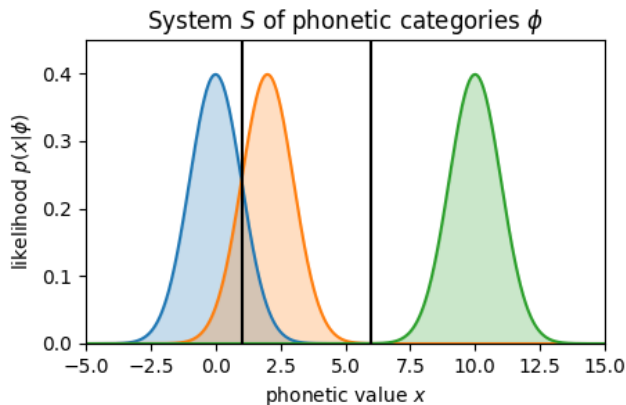
maximum-likelihood behavior

$$C(x) = \operatorname{argmax}_{\phi \in S} p(x|\phi)$$



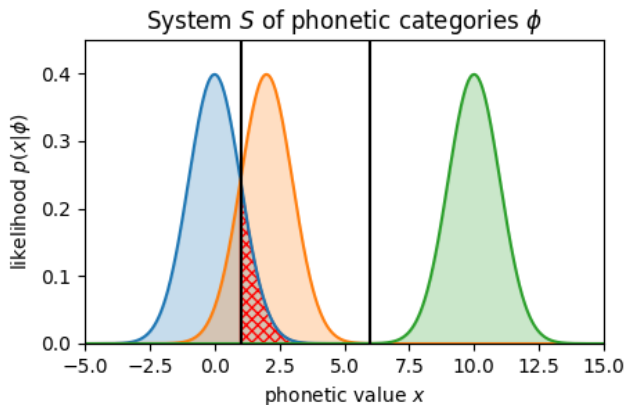
Optimal Listener Confusion

Tokens on 'wrong' side of boundaries
classified incorrectly



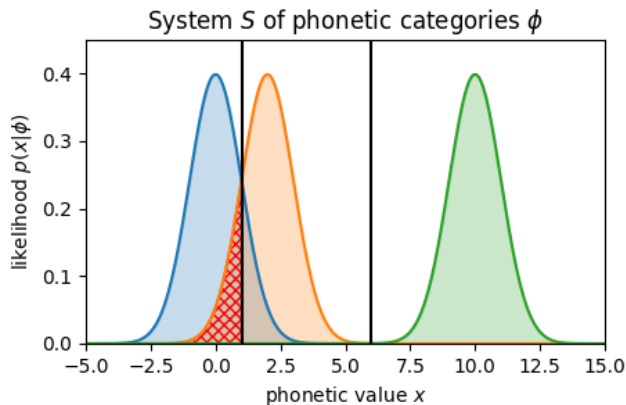
Optimal Listener Confusion

Tokens on 'wrong' side of boundaries
classified incorrectly



Optimal Listener Confusion

Tokens on 'wrong' side of boundaries
classified incorrectly



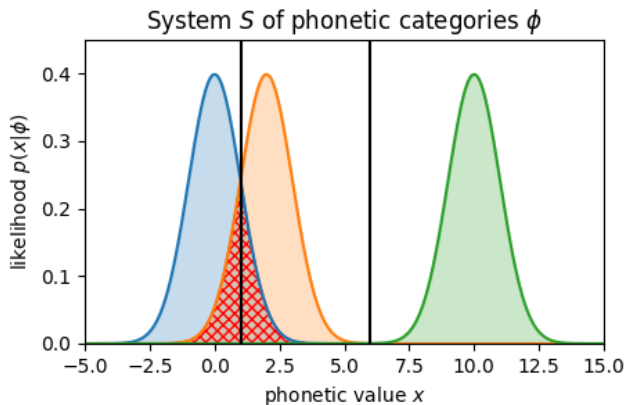
Optimal Listener Confusion

Tokens on 'wrong' side of boundaries classified incorrectly

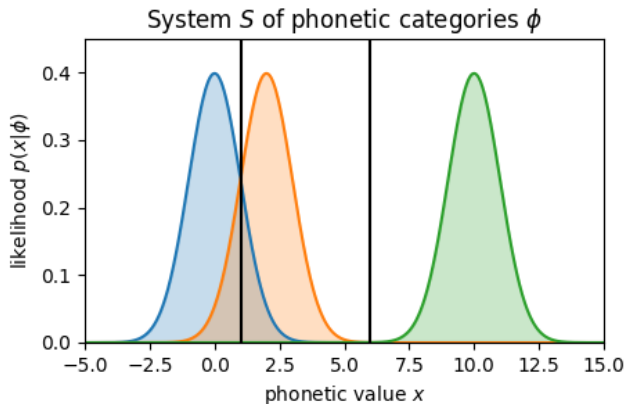
Confusion is equivalent to area of phonetic overlap (times normalizing constant)

classifier confusion rate

$$\frac{1}{|S|} \sum_{\phi \in S} \int p(x|\phi) P(C(x) \neq \phi) dx$$



Optimal Speaker Behavior



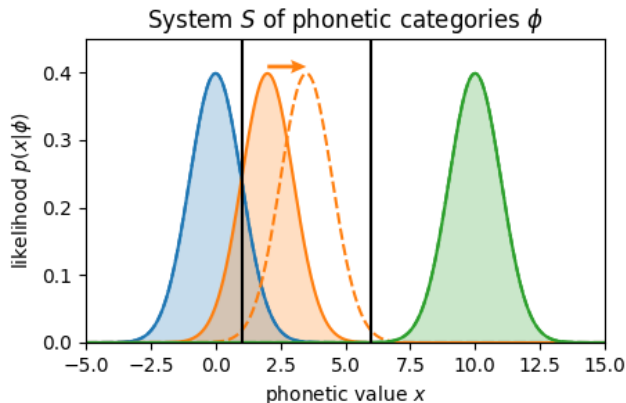
Optimal Speaker Behavior

A new set of distributions can reduce confusion, even if the listener expects the original distributions

least confusing system S'

Given set of possible distributions Φ :

$$S' = \left\{ \underset{\phi' \in \Phi}{\operatorname{argmin}} P(C(\phi') \neq \phi) \mid \phi \in S \right\}$$



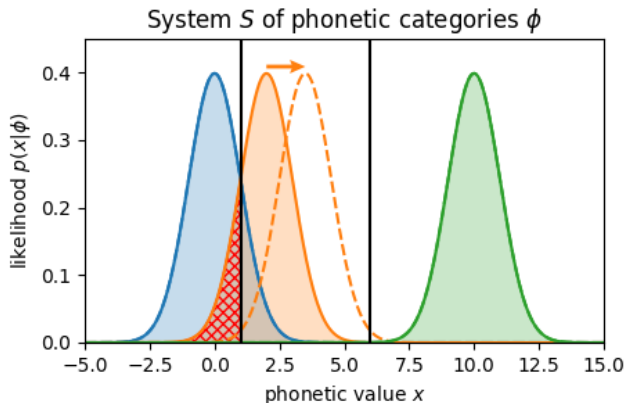
Optimal Speaker Behavior

A new set of distributions can reduce confusion, even if the listener expects the original distributions

least confusing system S'

Given set of possible distributions Φ :

$$S' = \left\{ \underset{\phi' \in \Phi}{\operatorname{argmin}} P(C(\phi') \neq \phi) \mid \phi \in S \right\}$$



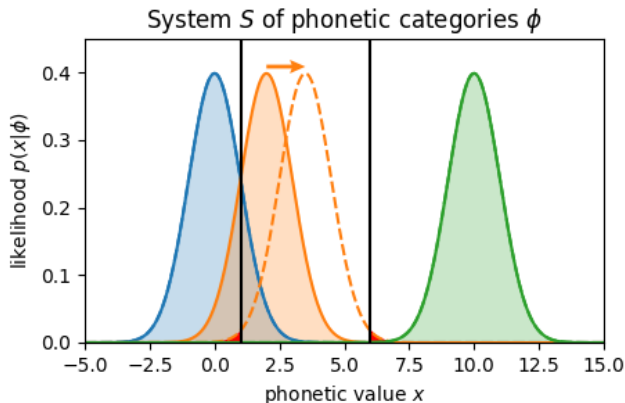
Optimal Speaker Behavior

A new set of distributions can reduce confusion, even if the listener expects the original distributions

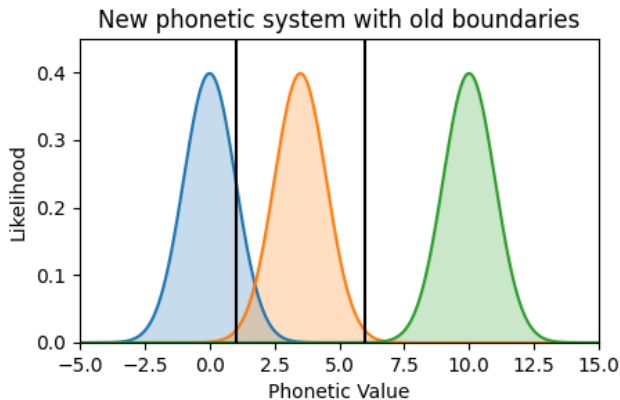
least confusing system S'

Given set of possible distributions Φ :

$$S' = \left\{ \underset{\phi' \in \Phi}{\operatorname{argmin}} P(C(\phi') \neq \phi) \mid \phi \in S \right\}$$

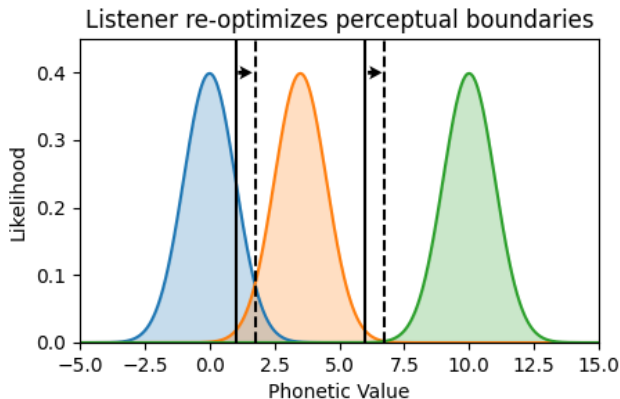


Iterated Confusion Minimization



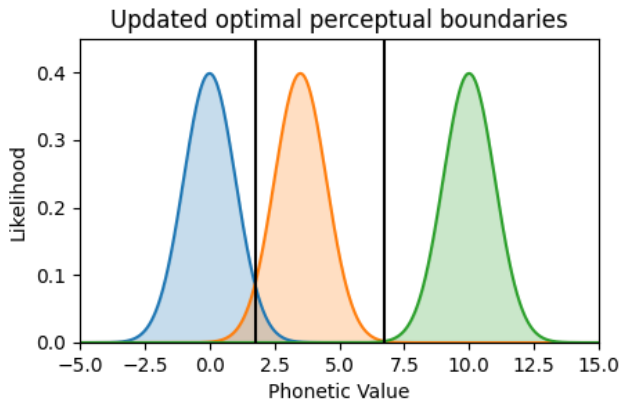
Iterated Confusion Minimization

Given speaker's new system, the optimal listener updates their perceptual boundaries



Iterated Confusion Minimization

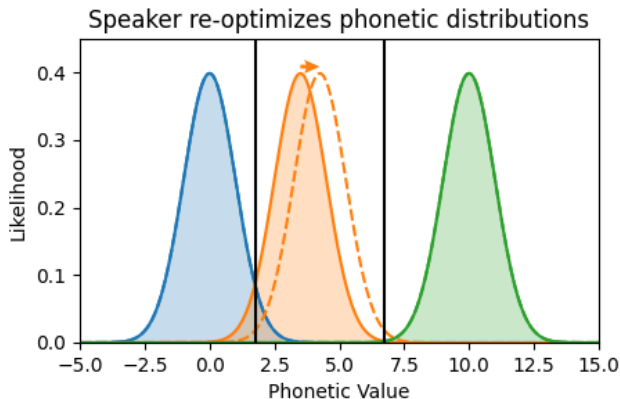
Given speaker's new system, the optimal listener updates their perceptual boundaries



Iterated Confusion Minimization

Given speaker's new system, the optimal listener updates their perceptual boundaries

Then, the optimal speaker updates their phonetic distributions again

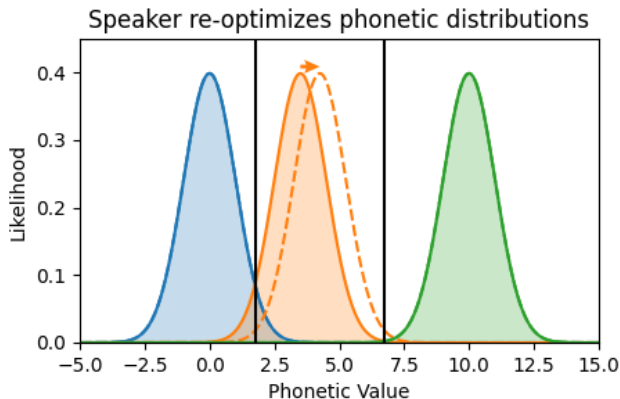


Iterated Confusion Minimization

Given speaker's new system, the optimal listener updates their perceptual boundaries

Then, the optimal speaker updates their phonetic distributions again

We will call this repeated process
Iterated Confusion Minimization (ICM)



Iterated Confusion Minimization Algorithm

Iterated Confusion Minimization (ICM)

Input: $S_0 \subseteq \Phi$, where Φ is set of possible phonetic distributions

Output: $S_i \subseteq \Phi$

1: $i \leftarrow 0$

2: **repeat**

3: ▷ *Optimal listener behavior*

4: $C_i(x) \leftarrow \operatorname{argmax}_{\phi \in S_i} p(x|\phi)$

5: ▷ *Optimal speaker behavior*

6: $S_{i+1} \leftarrow \{\operatorname{argmin}_{\phi' \in \Phi} P(C_i(\phi') \neq \phi) | \phi \in S_i\}$

7: $i \leftarrow i + 1$

8: **until** $S_i = S_{i-1}$

9: **return** S_i

ICM over the Vowel Space

McGahay (2025, under review) simulated ICM over the vowel space to predict typology

ICM over the Vowel Space

McGahay (2025, under review) simulated ICM over the vowel space to predict typology

Vowels modeled as bivariate normal distributions over $F1 \times F2$ space

ICM over the Vowel Space

McGahay (2025, under review) simulated ICM over the vowel space to predict typology

Vowels modeled as bivariate normal distributions over $F1 \times F2$ space

- Boundaries on vowel formant means adapted from Liljencrants and Lindblom (1972)

ICM over the Vowel Space

McGahay (2025, under review) simulated ICM over the vowel space to predict typology

Vowels modeled as bivariate normal distributions over $F1 \times F2$ space

- Boundaries on vowel formant means adapted from Liljencrants and Lindblom (1972)
- Formant standard deviations based on linear regression over cross-linguistic data set

ICM over the Vowel Space

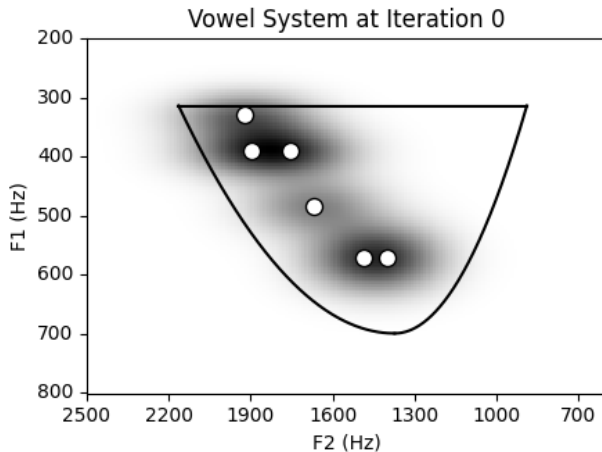
McGahay (2025, under review) simulated ICM over the vowel space to predict typology

Vowels modeled as bivariate normal distributions over $F1 \times F2$ space

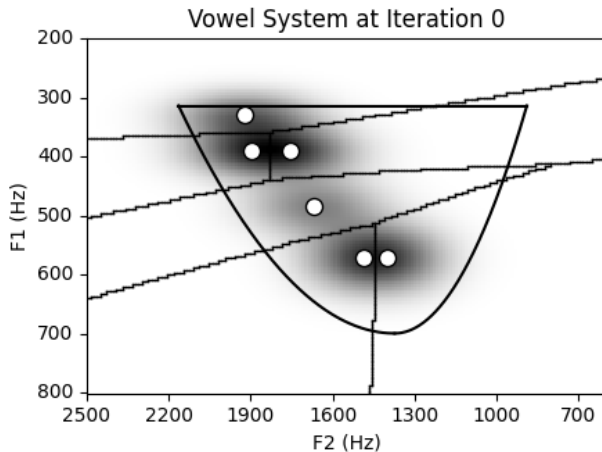
- Boundaries on vowel formant means adapted from Liljencrants and Lindblom (1972)
- Formant standard deviations based on linear regression over cross-linguistic data set

Outputs compared against a typological baseline based on Schwartz et al.'s (1997a) analysis of the UPSID (Maddieson, 1984)

Simulating Sound Change

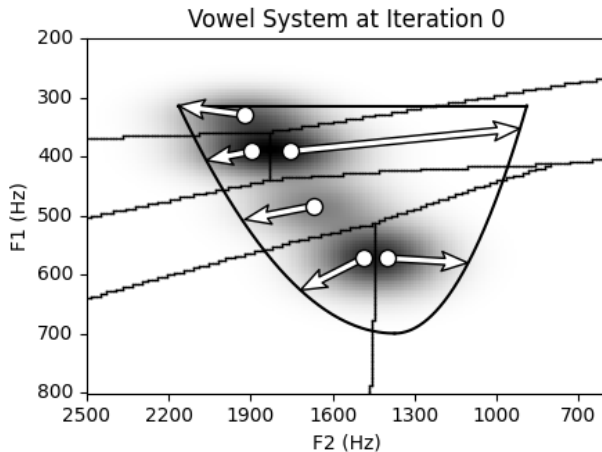


Simulating Sound Change



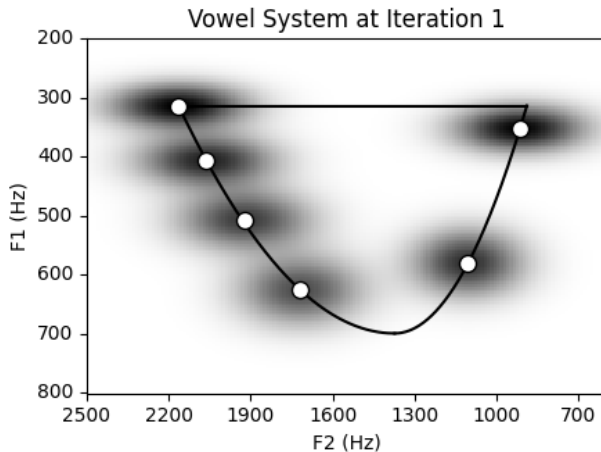
Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery



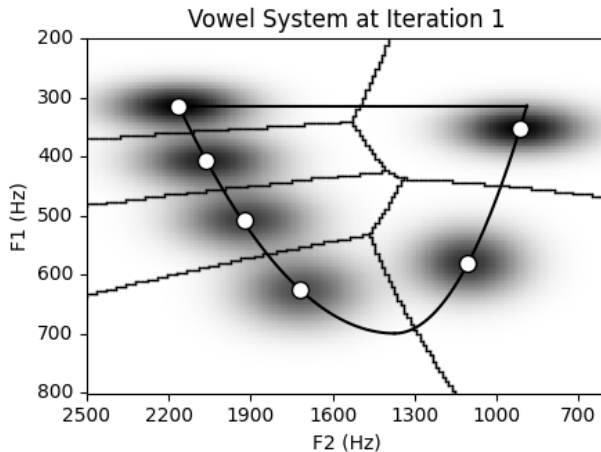
Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery



Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

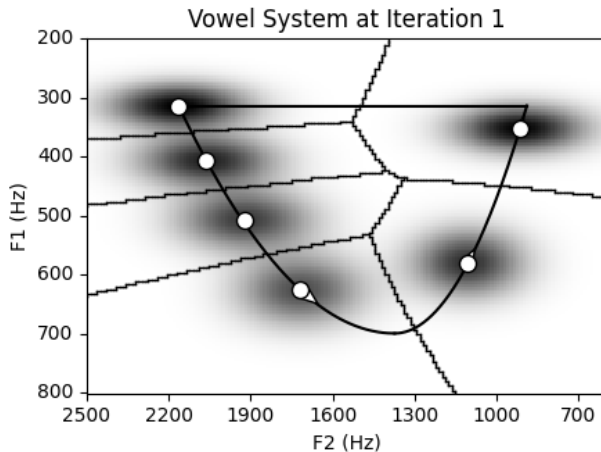


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

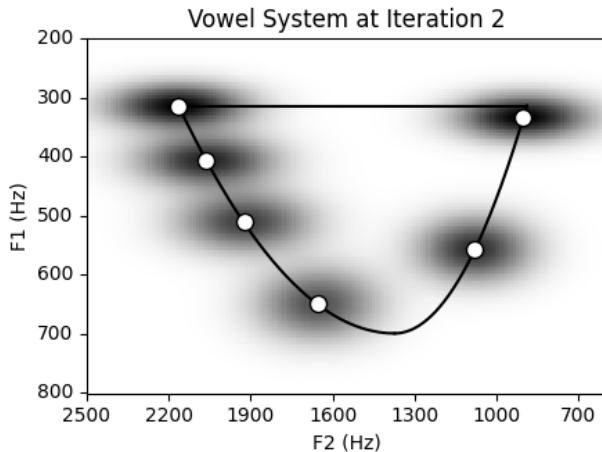


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

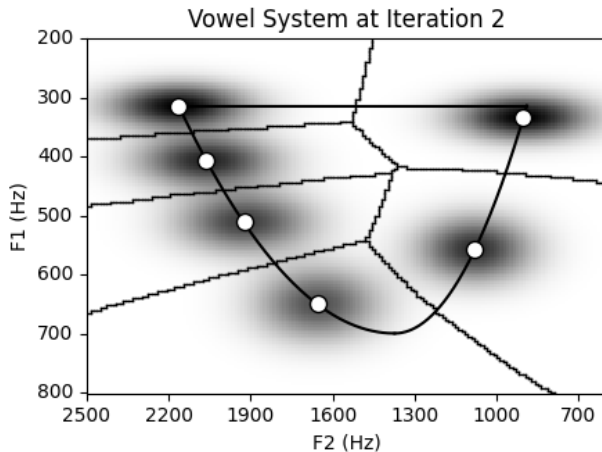


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

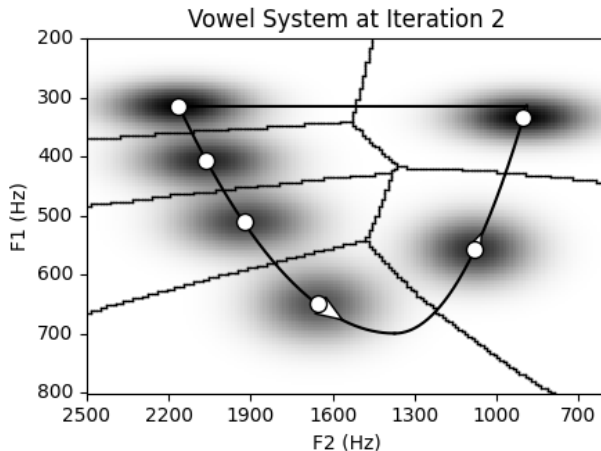


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

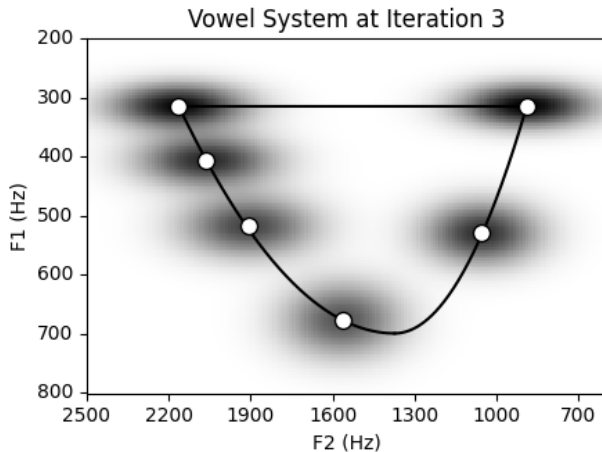


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

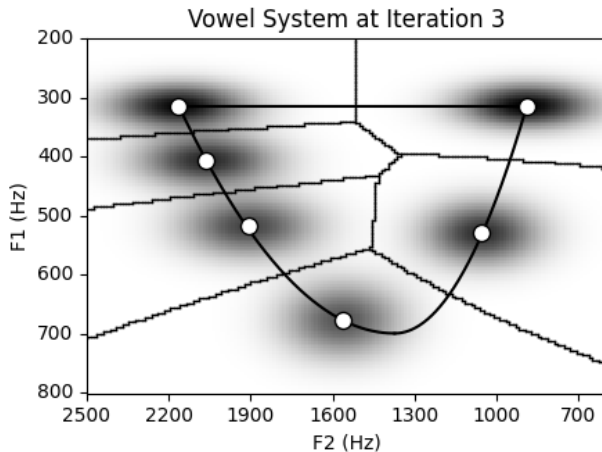


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

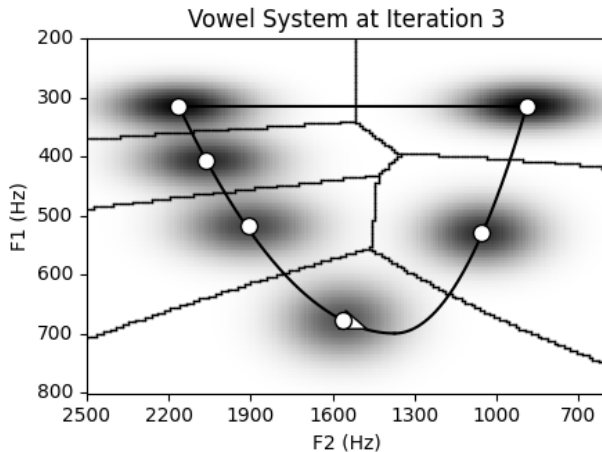


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

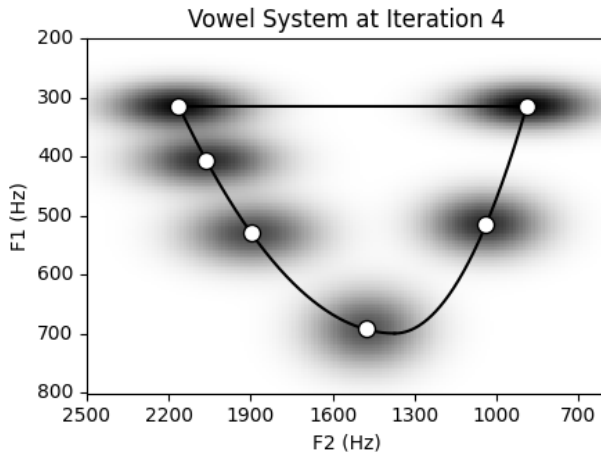


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

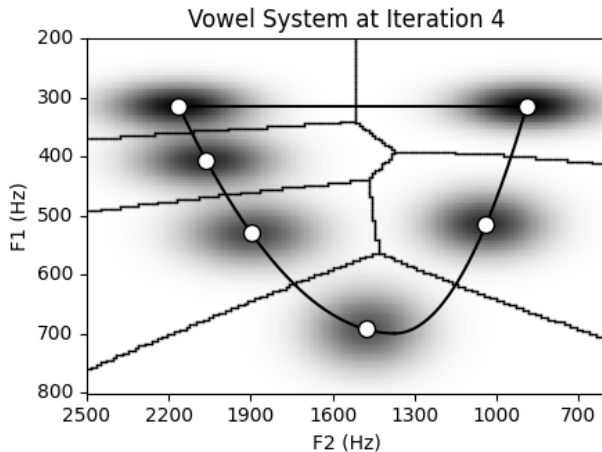


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

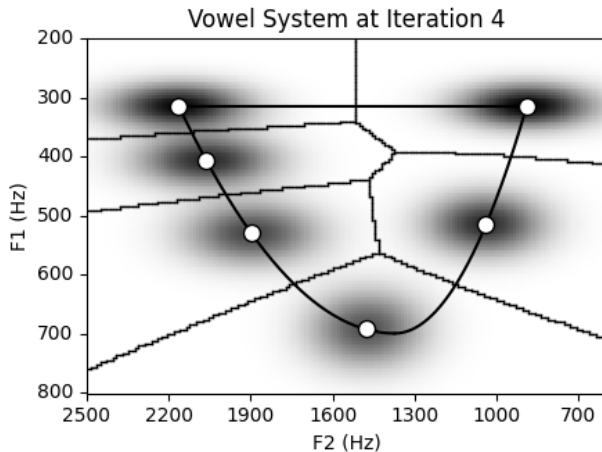


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

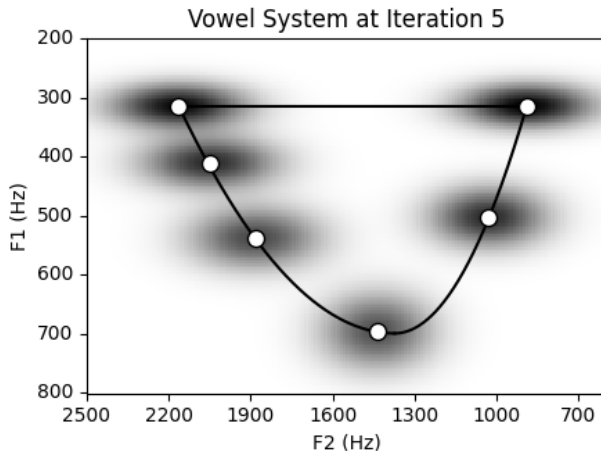


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

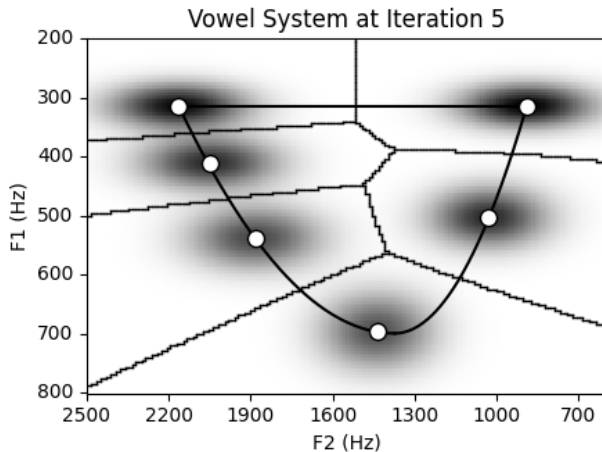


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

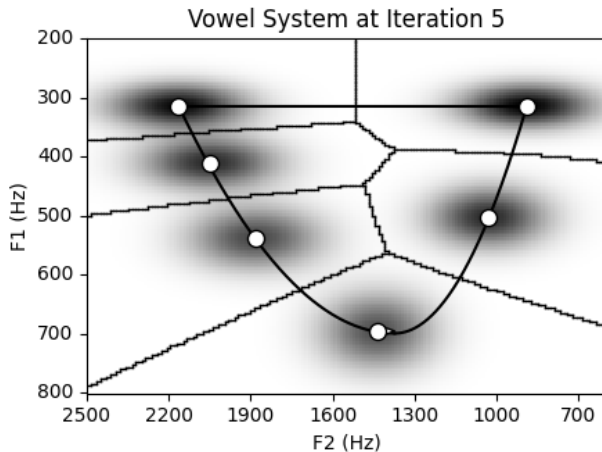


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

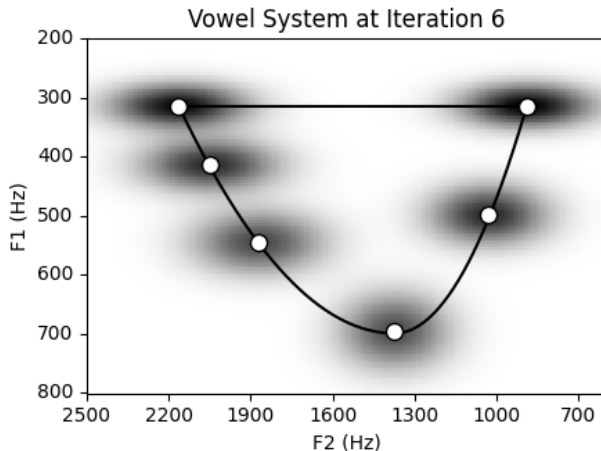


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

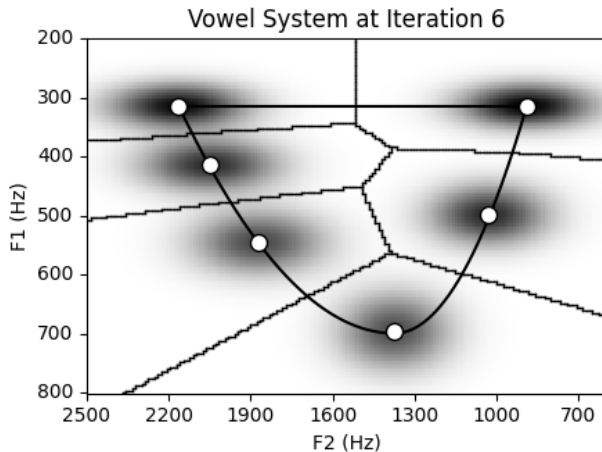


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

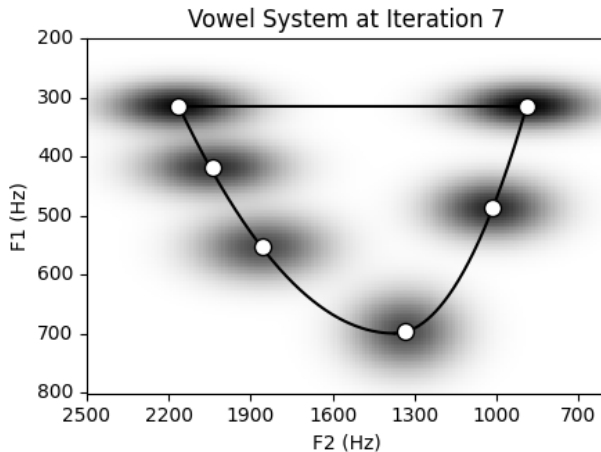


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

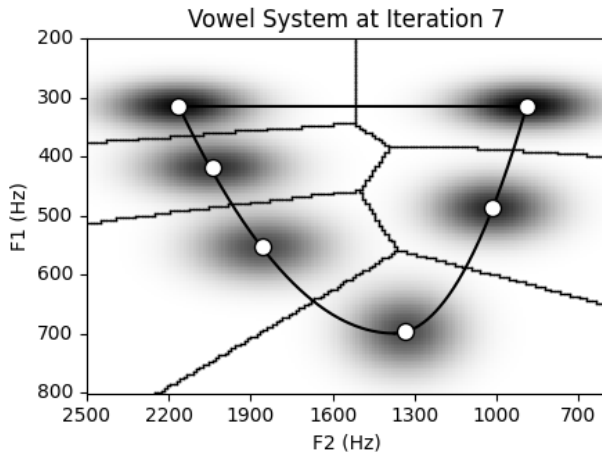


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

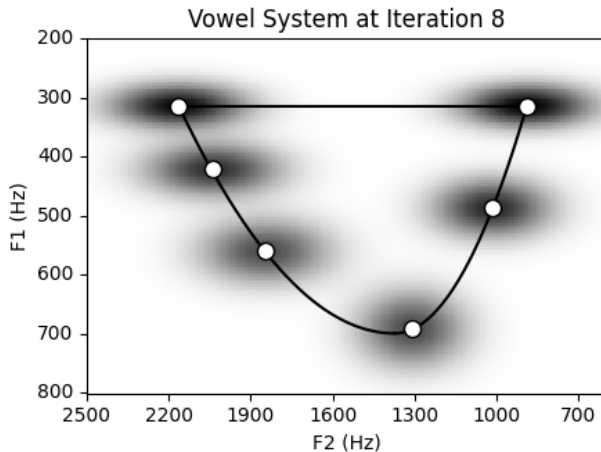


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

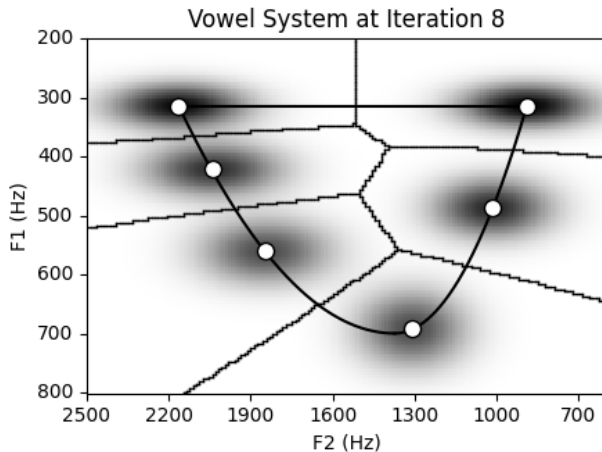


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

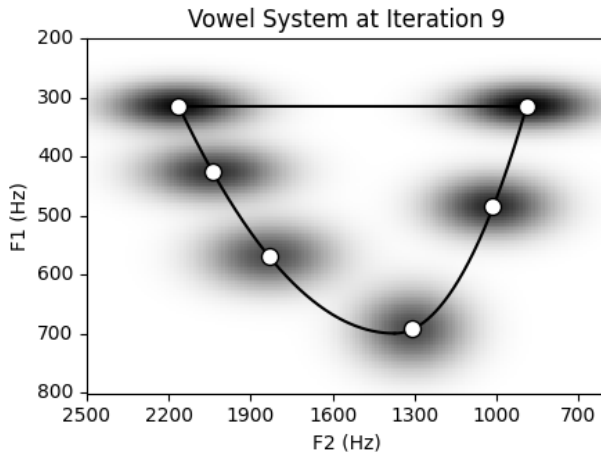


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

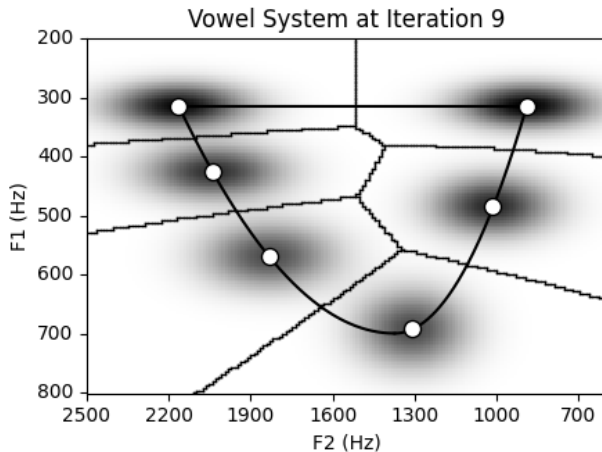


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

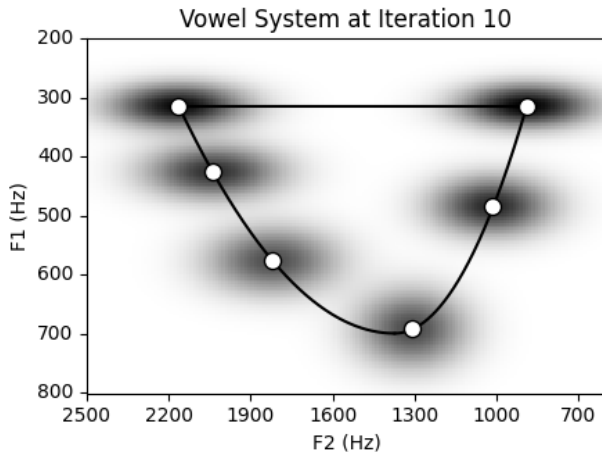


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

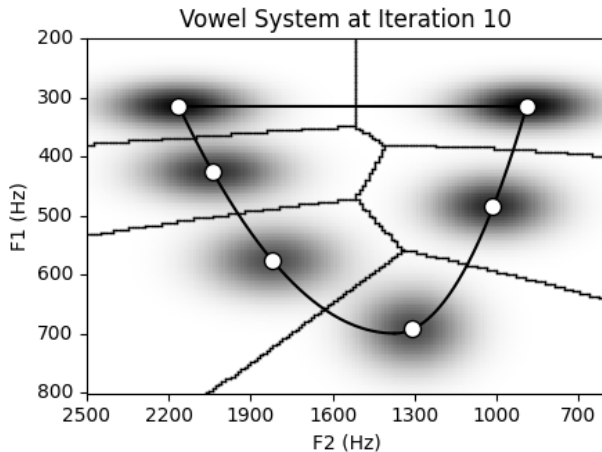


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

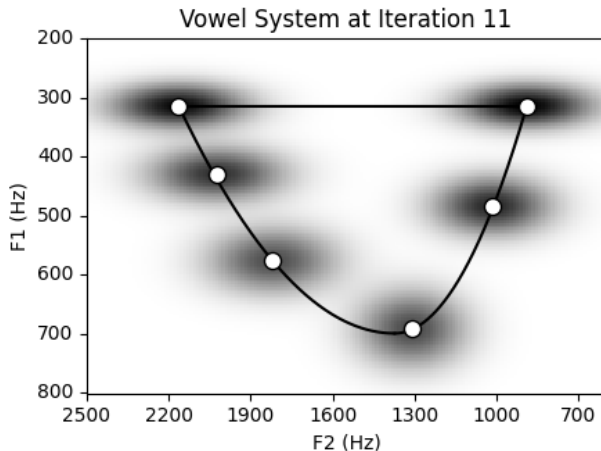


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

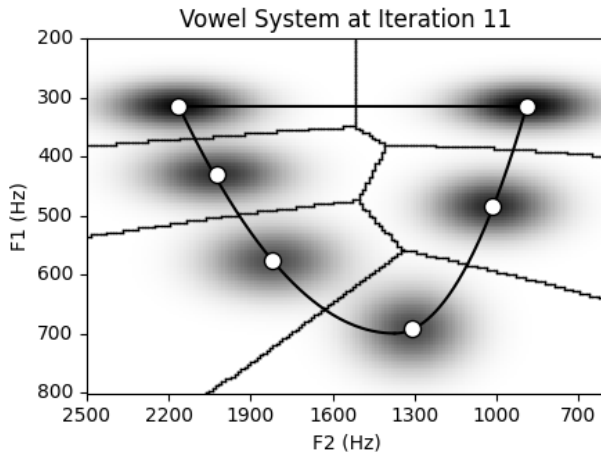


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

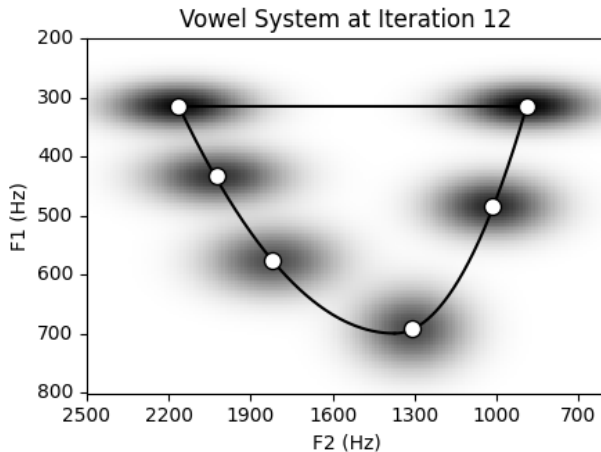


Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift



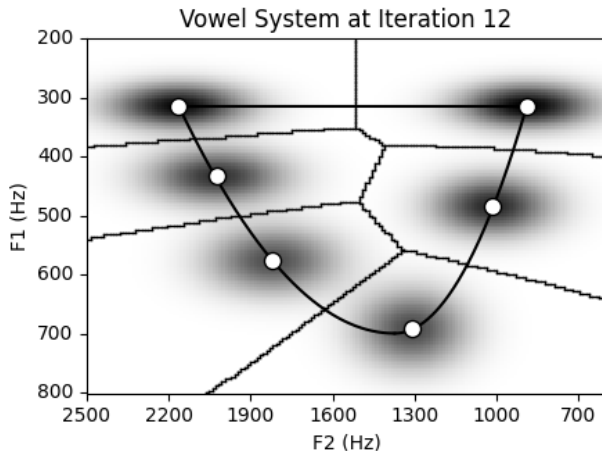
Simulating Sound Change

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

ICM reaches equilibrium with naturalistic, dispersed vowel system



Simulating Sound Change

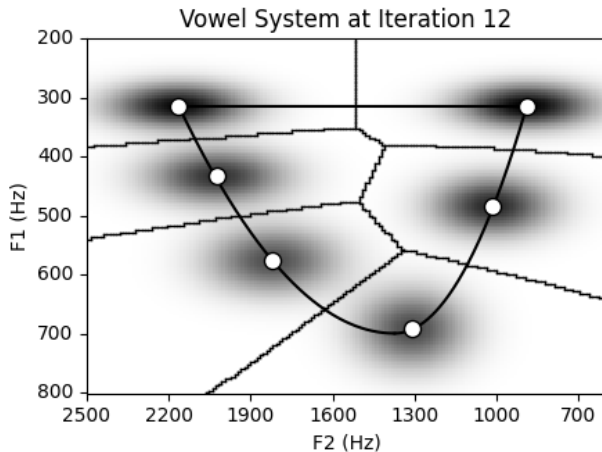
ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

ICM reaches equilibrium with naturalistic, dispersed vowel system

- /ieɛaou/ 2nd most common 6V system (Schwartz et al., 1997a)



Predicting Vowel System Typology

Every vowel system type in UPSID up to 12 vowels
is stable under some parameter setting

Predicting Vowel System Typology

Every vowel system type in UPSID up to 12 vowels
is stable under some parameter setting

- Past models only successful up to 7 or 8
(Schwartz et al., 1997b; de Boer, 1999)

Predicting Vowel System Typology

Every vowel system type in UPSID up to 12 vowels
is stable under some parameter setting

- Past models only successful up to 7 or 8
(Schwartz et al., 1997b; de Boer, 1999)

Variation predicted based on initial condition

Predicting Vowel System Typology

Every vowel system type in UPSID up to 12 vowels
is stable under some parameter setting

- Past models only successful up to 7 or 8
(Schwartz et al., 1997b; de Boer, 1999)

Variation predicted based on initial condition

- Equilibrium systems correspond to local optima

Predicting Vowel System Typology

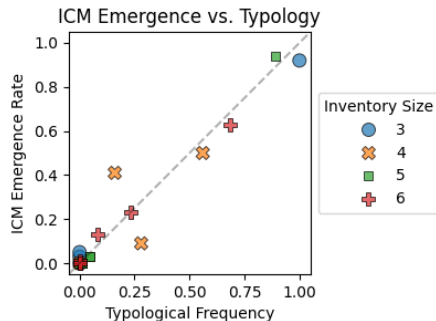
Every vowel system type in UPSID up to 12 vowels is stable under some parameter setting

- Past models only successful up to 7 or 8 (Schwartz et al., 1997b; de Boer, 1999)

Variation predicted based on initial condition

- Equilibrium systems correspond to local optima

Typological frequency up to 6 vowels (67% of languages) tightly correlated with ICM emergence rate from random initial system



Predicting Vowel System Typology

Every vowel system type in UPSID up to 12 vowels is stable under some parameter setting

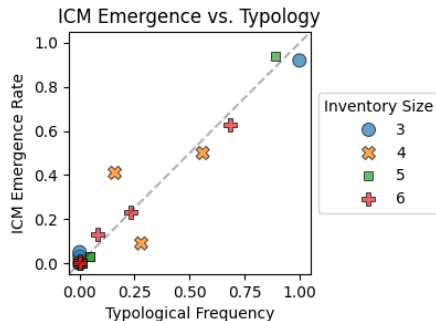
- Past models only successful up to 7 or 8 (Schwartz et al., 1997b; de Boer, 1999)

Variation predicted based on initial condition

- Equilibrium systems correspond to local optima

Typological frequency up to 6 vowels (67% of languages) tightly correlated with ICM emergence rate from random initial system

- ⇒ Typological frequency of vowel systems reflects how easily they emerge through sound change



Generalizing ICM to Phonetic Sequences

Generalizing ICM to Phonetic Sequences

Today: ICM over CV sequences

Generalizing ICM to Phonetic Sequences

Today: ICM over CV sequences

- simplifying assumption: consonant distributions held fixed (while vowel distributions vary)

Generalizing ICM to Phonetic Sequences

Today: ICM over CV sequences

- simplifying assumption: consonant distributions held fixed (while vowel distributions vary)

2 case studies:

Generalizing ICM to Phonetic Sequences

Today: ICM over CV sequences

- simplifying assumption: consonant distributions held fixed (while vowel distributions vary)

2 case studies:

- Vowel allophones adjacent to uvulars

Generalizing ICM to Phonetic Sequences

Today: ICM over CV sequences

- simplifying assumption: consonant distributions held fixed (while vowel distributions vary)

2 case studies:

- Vowel allophones adjacent to uvulars
- Vertical vowel systems

Maximum-Likelihood vs. Posterior-Tracking Listener Behavior

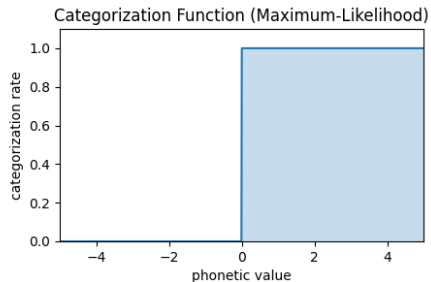
maximum-likelihood categorization

$$C(x) = \operatorname{argmax}_{\phi \in S} p(x|\phi)$$

Maximum-Likelihood vs. Posterior-Tracking Listener Behavior

maximum-likelihood categorization

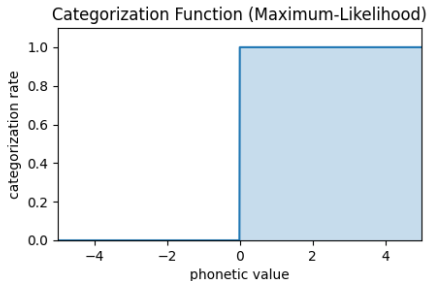
$$C(x) = \operatorname{argmax}_{\phi \in S} p(x|\phi)$$



Maximum-Likelihood vs. Posterior-Tracking Listener Behavior

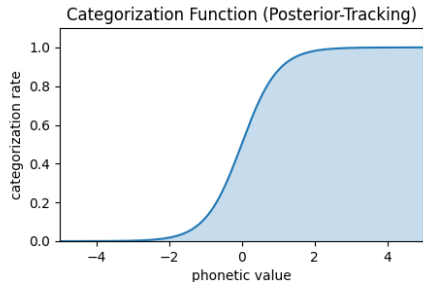
maximum-likelihood categorization

$$C(x) = \operatorname{argmax}_{\phi \in S} p(x|\phi)$$



posterior-tracking categorization

$$P(C(x) = \phi) = P(\phi|x) = \frac{p(x|\phi)P(\phi)}{p(x)}$$



Human speech perception resembles posterior-tracking behavior (cf. Clayards et al., 2008)

Posterior-Tracking Categorization of Sequences

Posterior-Tracking Categorization of Sequences

With some idealizing independence assumptions, a posterior-tracking classifier C categorizes a CV token x as CV type ϕ at the following rate:

$$P(C(x) = \phi) = P(\phi|x) = \frac{p(x_c|\phi_c)p(x_v|\phi_v)P(\phi)}{\sum_{\phi' \in \mathcal{S}} p(x_c|\phi'_c)p(x_v|\phi'_v)P(\phi')}$$

Posterior-Tracking Categorization of Sequences

With some idealizing independence assumptions, a posterior-tracking classifier C categorizes a CV token x as CV type ϕ at the following rate:

$$P(C(x) = \phi) = P(\phi|x) = \frac{p(x_c|\phi_c)p(x_v|\phi_v)P(\phi)}{\sum_{\phi' \in \mathcal{S}} p(x_c|\phi'_c)p(x_v|\phi'_v)P(\phi')}$$

where:

- $p(x_v|\phi_v)$: vowel likelihoods based on existing vowel distribution model (McGahay, 2025)

Posterior-Tracking Categorization of Sequences

With some idealizing independence assumptions, a posterior-tracking classifier C categorizes a CV token x as CV type ϕ at the following rate:

$$P(C(x) = \phi) = P(\phi|x) = \frac{p(x_c|\phi_c)p(x_v|\phi_v)P(\phi)}{\sum_{\phi' \in \mathcal{S}} p(x_c|\phi'_c)p(x_v|\phi'_v)P(\phi')}$$

where:

- $p(x_v|\phi_v)$: vowel likelihoods based on existing vowel distribution model (McGahay, 2025)
- $P(\phi)$: prior probabilities over sequence types (cancel out assuming uniform priors)

Posterior-Tracking Categorization of Sequences

With some idealizing independence assumptions, a posterior-tracking classifier C categorizes a CV token x as CV type ϕ at the following rate:

$$P(C(x) = \phi) = P(\phi|x) = \frac{p(x_c|\phi_c)p(x_v|\phi_v)P(\phi)}{\sum_{\phi' \in \mathcal{S}} p(x_c|\phi'_c)p(x_v|\phi'_v)P(\phi')}$$

where:

- $p(x_v|\phi_v)$: vowel likelihoods based on existing vowel distribution model (McGahay, 2025)
- $P(\phi)$: prior probabilities over sequence types (cancel out assuming uniform priors)
- $p(x_c|\phi_c)$: consonant likelihoods (difficult to explicitly model)

Posterior-Tracking Categorization of Sequences

With some idealizing independence assumptions, a posterior-tracking classifier C categorizes a CV token x as CV type ϕ at the following rate:

$$P(C(x) = \phi) = P(\phi|x) = \frac{p(x_c|\phi_c)p(x_v|\phi_v)P(\phi)}{\sum_{\phi' \in \mathcal{S}} p(x_c|\phi'_c)p(x_v|\phi'_v)P(\phi')}$$

where:

- $p(x_v|\phi_v)$: vowel likelihoods based on existing vowel distribution model (McGahay, 2025)
- $P(\phi)$: prior probabilities over sequence types (cancel out assuming uniform priors)
- $p(x_c|\phi_c)$: consonant likelihoods (difficult to explicitly model)
 - Possible to select values that emulate expected consonant confusability, e.g. based on experimental confusion matrices (Miller and Nicely, 1955)

ICM generalized over CV sequences

ICM over CV sequences with fixed consonant distributions

Input: $S_0 \subseteq \Phi_c \times \Phi_v$: initial system of CV sequences $\phi = \phi_c \phi_v$, $\phi_c \in \Phi_c$, $\phi_v \in \Phi_v$, where Φ_c and Φ_v denote sets of possible consonant and vowel distributions.

Output: $S_i \subseteq \Phi_c \times \Phi_v$: the sound system S_i resulting from i iterations over S_0

1: $i \leftarrow 0$

2: **repeat**

3: ▷ *Optimal listener behavior*

4: $P(C_i(x) = \phi) = P_{S_i}(\phi|x)$

5: ▷ *Optimal speaker behavior*

6: $S_{i+1} = \{\text{argmin}_{\phi' \in \{\phi_c\} \times \Phi_v} P(C_i(\phi') \neq \phi) \mid \phi \in S_i\}$

7: $i \leftarrow i + 1$

8: **until** $S_i = S_{i-1}$

9: **return** S_i

Case Study 1: Allophony Adjacent to Uvulars

Vowels commonly lowered/retracted adjacent to uvulars, e.g.:

- Arabic (McCarthy, 1994)
- Quechua (Gallagher, 2016)
- West Greenlandic (Rischel, 1974)

Case Study 1: Allophony Adjacent to Uvulars

Vowels commonly lowered/retracted adjacent to uvulars, e.g.:

- Arabic (McCarthy, 1994)
- Quechua (Gallagher, 2016)
- West Greenlandic (Rischel, 1974)

ICM simulation:

- 6 sequences {/ki/, /ka/, /ku/, /qi/, /qa/, /qu/}

Case Study 1: Allophony Adjacent to Uvulars

Vowels commonly lowered/retracted adjacent to uvulars, e.g.:

- Arabic (McCarthy, 1994)
- Quechua (Gallagher, 2016)
- West Greenlandic (Rischel, 1974)

ICM simulation:

- 6 sequences {/ki/, /ka/, /ku/, /qi/, /qa/, /qu/}
- Small differences in mean vowel formants ($\leq 20\text{Hz}$) due to coarticulation

Case Study 1: Allophony Adjacent to Uvulars

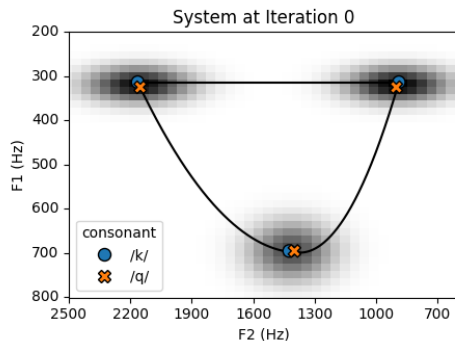
Vowels commonly lowered/retracted adjacent to uvulars, e.g.:

- Arabic (McCarthy, 1994)
- Quechua (Gallagher, 2016)
- West Greenlandic (Rischel, 1974)

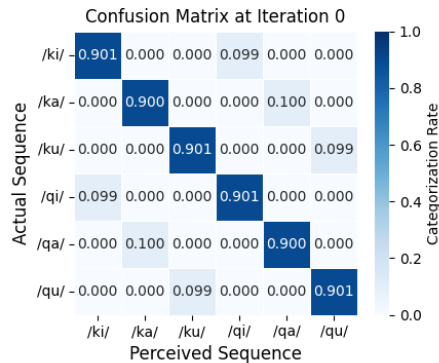
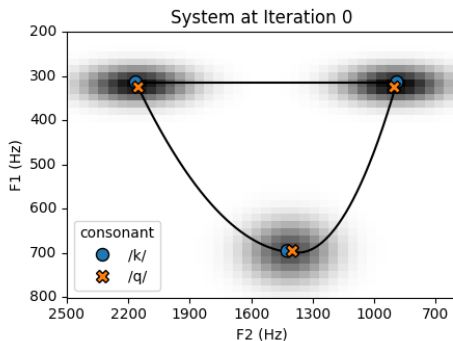
ICM simulation:

- 6 sequences {/ki/, /ka/, /ku/, /qi/, /qa/, /qu/}
- Small differences in mean vowel formants ($\leq 20\text{Hz}$) due to coarticulation
- 10% of /k/ and /q/ tokens confused without disambiguation from vowel

Case Study 1: Allophony Adjacent to Uvulars

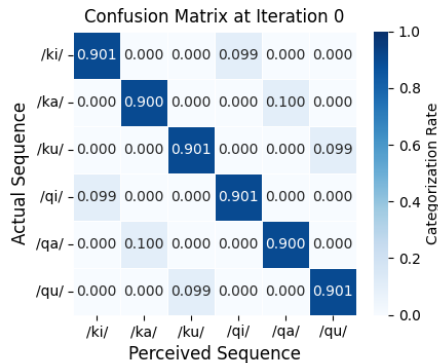
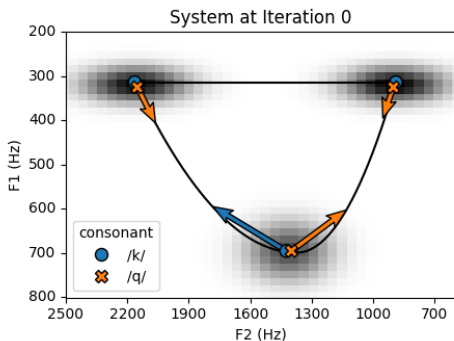


Case Study 1: Allophony Adjacent to Uvulars



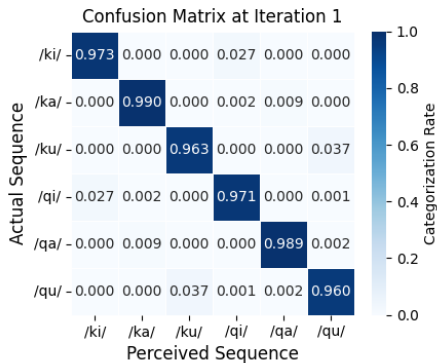
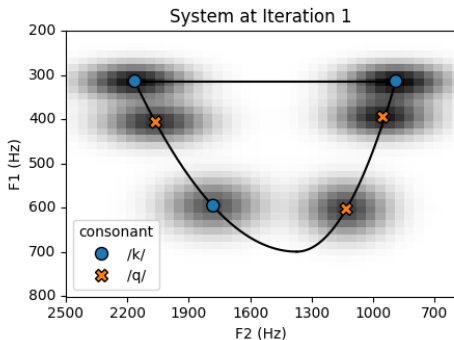
Case Study 1: Allophony Adjacent to Uvulars

Initial iteration amplifies small differences attributable to coarticulation into robust allophones



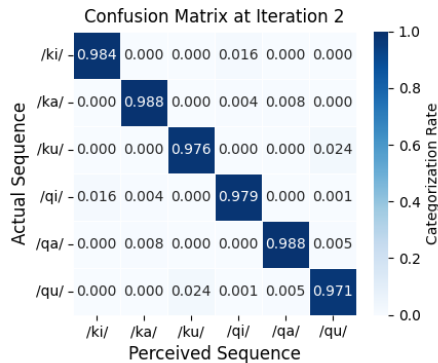
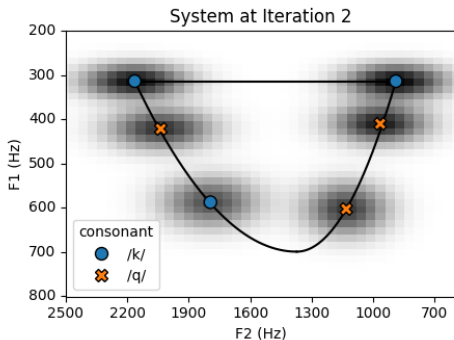
Case Study 1: Allophony Adjacent to Uvulars

Initial iteration amplifies small differences attributable to coarticulation into robust allophones



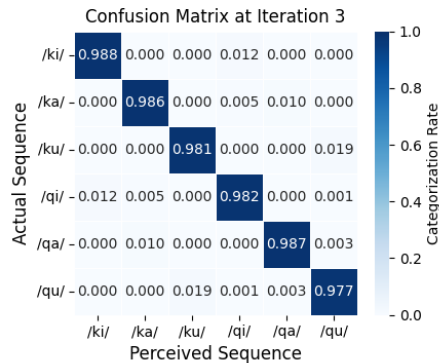
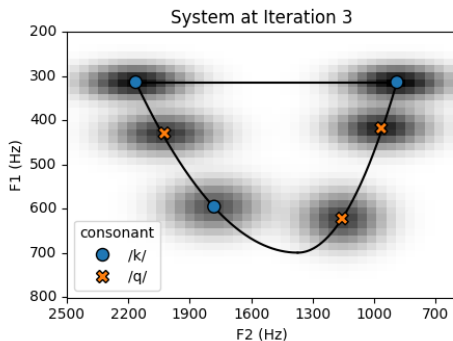
Case Study 1: Allophony Adjacent to Uvulars

Initial iteration amplifies small differences attributable to coarticulation into robust allophones



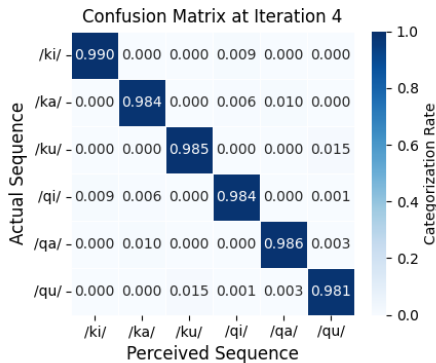
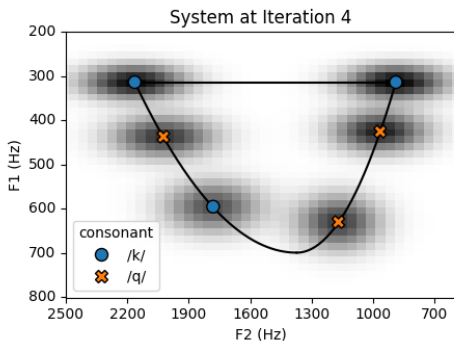
Case Study 1: Allophony Adjacent to Uvulars

Initial iteration amplifies small differences attributable to coarticulation into robust allophones



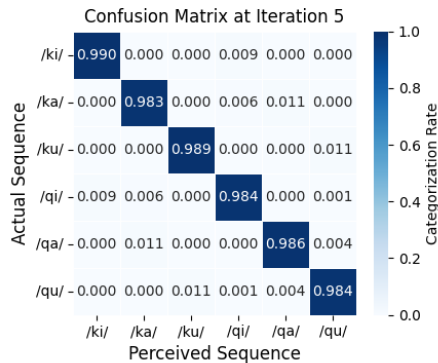
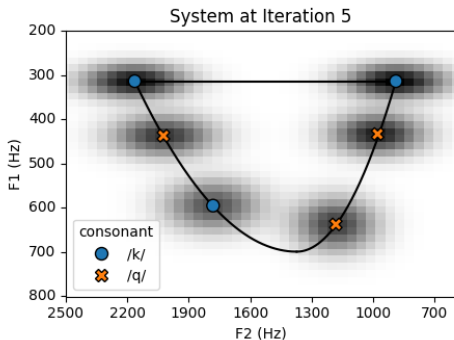
Case Study 1: Allophony Adjacent to Uvulars

Initial iteration amplifies small differences attributable to coarticulation into robust allophones



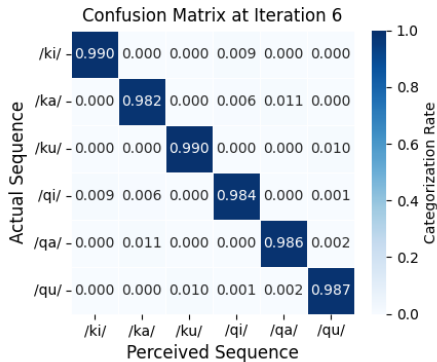
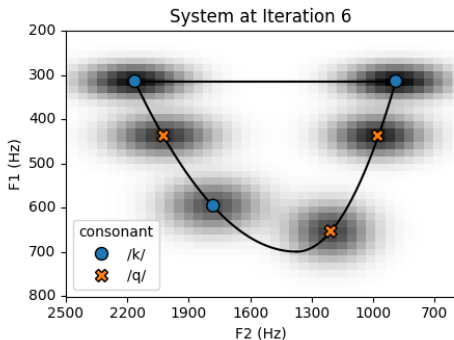
Case Study 1: Allophony Adjacent to Uvulars

Initial iteration amplifies small differences attributable to coarticulation into robust allophones



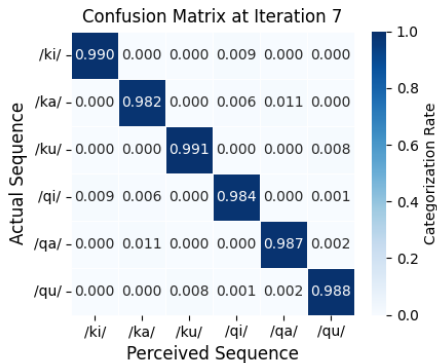
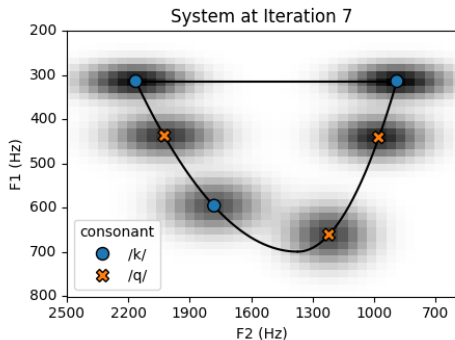
Case Study 1: Allophony Adjacent to Uvulars

Initial iteration amplifies small differences attributable to coarticulation into robust allophones



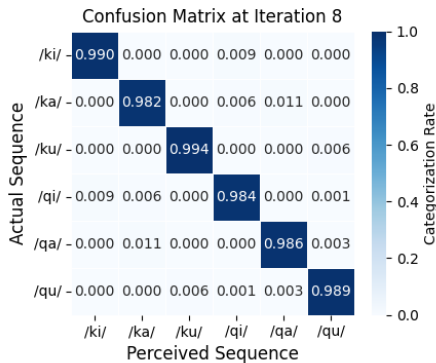
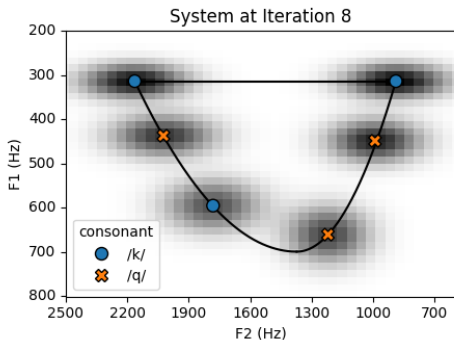
Case Study 1: Allophony Adjacent to Uvulars

Initial iteration amplifies small differences attributable to coarticulation into robust allophones



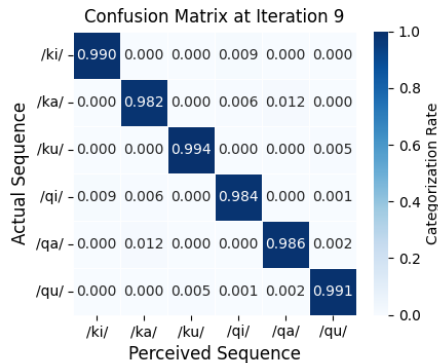
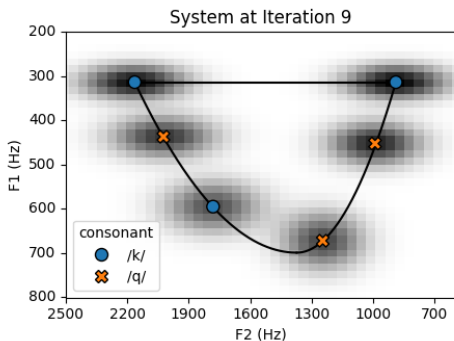
Case Study 1: Allophony Adjacent to Uvulars

Initial iteration amplifies small differences attributable to coarticulation into robust allophones



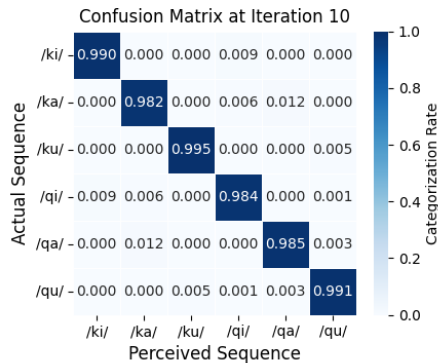
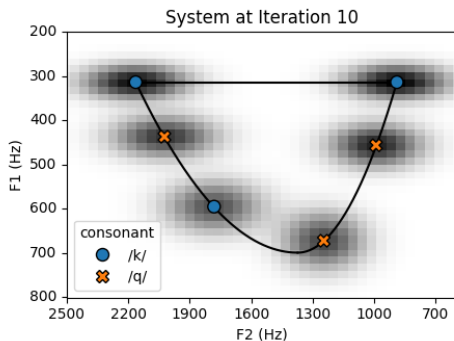
Case Study 1: Allophony Adjacent to Uvulars

Initial iteration amplifies small differences attributable to coarticulation into robust allophones



Case Study 1: Allophony Adjacent to Uvulars

Initial iteration amplifies small differences attributable to coarticulation into robust allophones



Equilibrium has naturalistic allophones [iæu] beside /k/ and [eao] beside /q/

Case Study 1: Allophony Adjacent to Uvulars

Vowel allophones reduce confusion by cuing difficult /k-q/ contrast

Case Study 1: Allophony Adjacent to Uvulars

Vowel allophones reduce confusion by cuing difficult /k-q/ contrast

- Consistent with Quechua speech perception experiments (Gallagher, 2016)

Case Study 1: Allophony Adjacent to Uvulars

Vowel allophones reduce confusion by cuing difficult /k-q/ contrast

- Consistent with Quechua speech perception experiments (Gallagher, 2016)

Further typological generalizations:

Case Study 1: Allophony Adjacent to Uvulars

Vowel allophones reduce confusion by cuing difficult /k-q/ contrast

- Consistent with Quechua speech perception experiments (Gallagher, 2016)

Further typological generalizations:

- Dramatic allophony more common in small vowel inventories

Case Study 1: Allophony Adjacent to Uvulars

Vowel allophones reduce confusion by cuing difficult /k-q/ contrast

- Consistent with Quechua speech perception experiments (Gallagher, 2016)

Further typological generalizations:

- Dramatic allophony more common in small vowel inventories
 - Allophony only perceptually optimizing in non-crowded spaces?

Case Study 1: Allophony Adjacent to Uvulars

Vowel allophones reduce confusion by cuing difficult /k-q/ contrast

- Consistent with Quechua speech perception experiments (Gallagher, 2016)

Further typological generalizations:

- Dramatic allophony more common in small vowel inventories
 - Allophony only perceptually optimizing in non-crowded spaces?
- Phonemic /k-q/ contrast less common in languages with crowded vowel inventories

Case Study 1: Allophony Adjacent to Uvulars

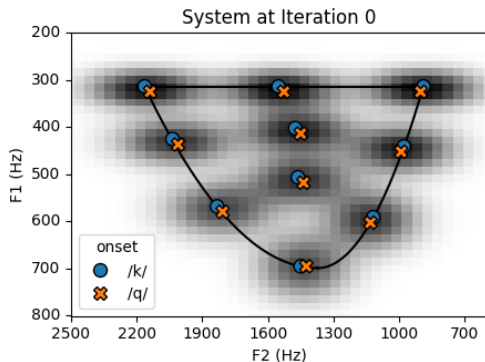
Vowel allophones reduce confusion by cuing difficult /k-q/ contrast

- Consistent with Quechua speech perception experiments (Gallagher, 2016)

Further typological generalizations:

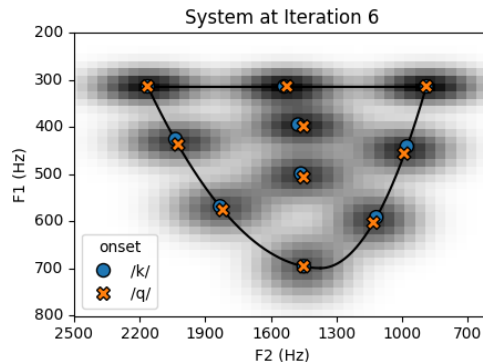
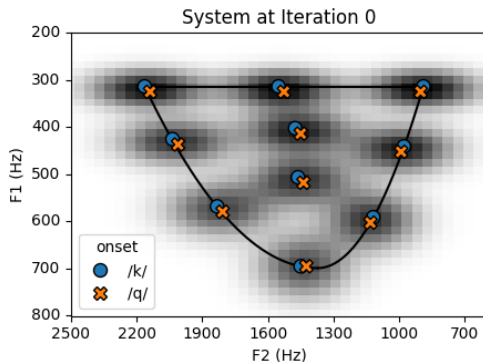
- Dramatic allophony more common in small vowel inventories
 - Allophony only perceptually optimizing in non-crowded spaces?
- Phonemic /k-q/ contrast less common in languages with crowded vowel inventories
 - /k-q/ contrast difficult to learn/maintain if not reinforced by allophones?

Case Study 1: Allophony Adjacent to Uvulars



Case Study 1: Allophony Adjacent to Uvulars

Allophony adjacent to uvulars not predicted for crowded vowel inventories



Case Study 2: Vertical Vowel Systems

Case Study 2: Vertical Vowel Systems

Uncommon but attested, e.g. /ɯɬa/ in Marshallese (Choi, 1992)

Case Study 2: Vertical Vowel Systems

Uncommon but attested, e.g. /ɯɬa/ in Marshallese (Choi, 1992)

- Under-utilization of F2 problematic for Dispersion Theory (cf. Vaux and Samuels, 2015)

Case Study 2: Vertical Vowel Systems

Uncommon but attested, e.g. /ɯɬa/ in Marshallese (Choi, 1992)

- Under-utilization of F2 problematic for Dispersion Theory (cf. Vaux and Samuels, 2015)

Front and back allophones adjacent to palatalized and labialized/velarized consonants

Case Study 2: Vertical Vowel Systems

Uncommon but attested, e.g. /ɯɬa/ in Marshallese (Choi, 1992)

- Under-utilization of F2 problematic for Dispersion Theory (cf. Vaux and Samuels, 2015)

Front and back allophones adjacent to palatalized and labialized/velarized consonants

- Consistent with dispersion if allophones considered?

Case Study 2: Vertical Vowel Systems

Uncommon but attested, e.g. /ɯɬa/ in Marshallese (Choi, 1992)

- Under-utilization of F2 problematic for Dispersion Theory (cf. Vaux and Samuels, 2015)

Front and back allophones adjacent to palatalized and labialized/velarized consonants

- Consistent with dispersion if allophones considered?

ICM simulation:

- 9 sequences of /ɪəa/ across palatalized /C^j/, plain /C/, and labialized /C^w/ contexts:
 - {/C^jɪ/, /C^jə/, /C^ja/, /Cɪ/, /Cə/, /Ca/, /C^wɪ/, /C^wə/, /C^wa/}

Case Study 2: Vertical Vowel Systems

Uncommon but attested, e.g. /ɯɬa/ in Marshallese (Choi, 1992)

- Under-utilization of F2 problematic for Dispersion Theory (cf. Vaux and Samuels, 2015)

Front and back allophones adjacent to palatalized and labialized/velarized consonants

- Consistent with dispersion if allophones considered?

ICM simulation:

- 9 sequences of /iəa/ across palatalized /C^j/, plain /C/, and labialized /C^w/ contexts:
 - {/C^ji/, /C^jə/, /C^ja/, /Ci/, /Cə/, /Ca/, /C^wi/, /C^wə/, /C^wa/}
- Small differences in mean F2 ($\pm 25\text{Hz}$) due to coarticulation

Case Study 2: Vertical Vowel Systems

Uncommon but attested, e.g. /ɯla/ in Marshallese (Choi, 1992)

- Under-utilization of F2 problematic for Dispersion Theory (cf. Vaux and Samuels, 2015)

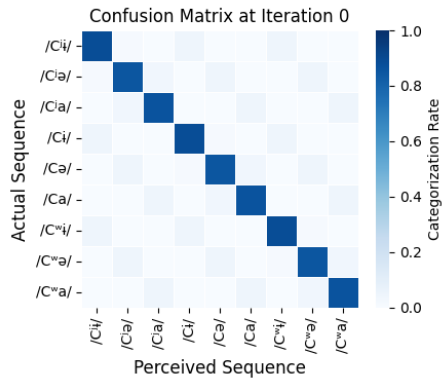
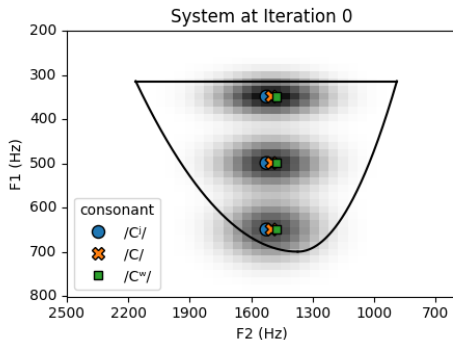
Front and back allophones adjacent to palatalized and labialized/velarized consonants

- Consistent with dispersion if allophones considered?

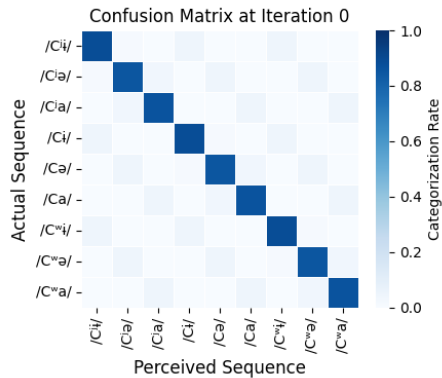
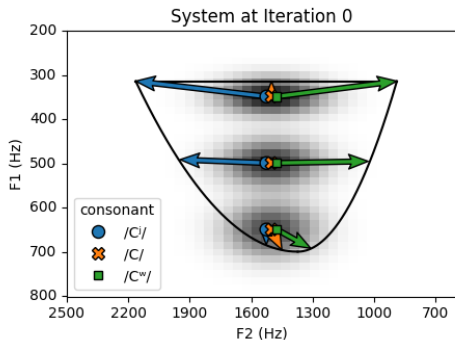
ICM simulation:

- 9 sequences of /iəa/ across palatalized /C^j/, plain /C/, and labialized /C^w/ contexts:
 - {/C^ji/, /C^jə/, /C^ja/, /Ci/, /Cə/, /Ca/, /C^wi/, /C^wə/, /C^wa/}
- Small differences in mean F2 ($\pm 25\text{Hz}$) due to coarticulation
- 10% of consonant tokens confused without disambiguation from vowel

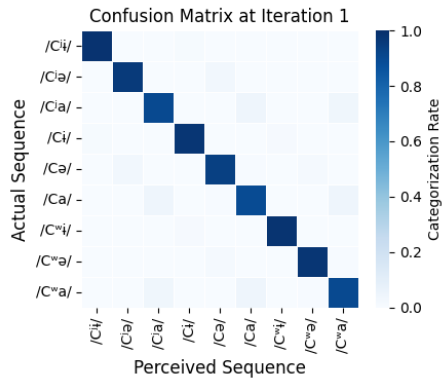
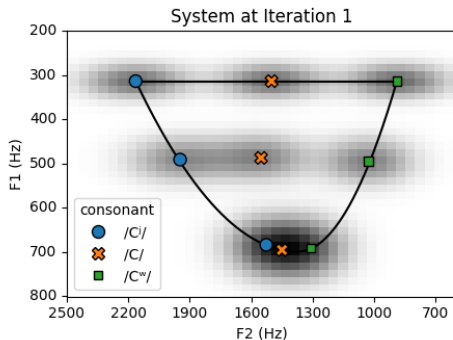
Case Study 2: Vertical Vowel Systems



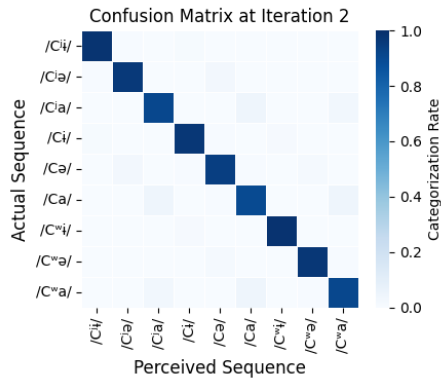
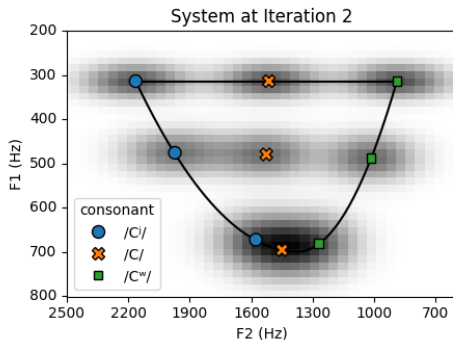
Case Study 2: Vertical Vowel Systems



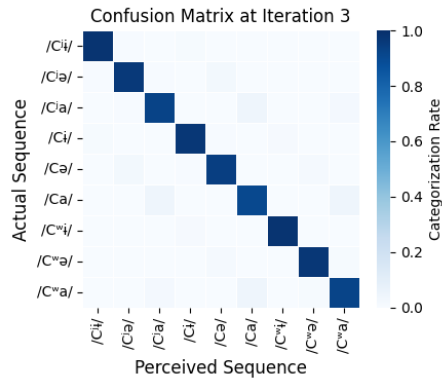
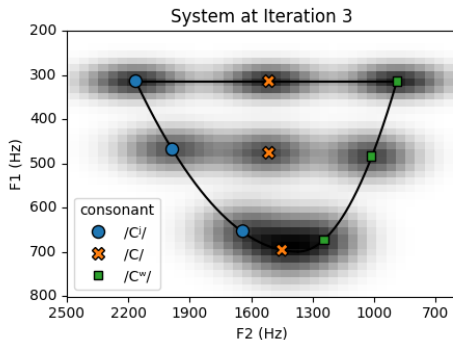
Case Study 2: Vertical Vowel Systems



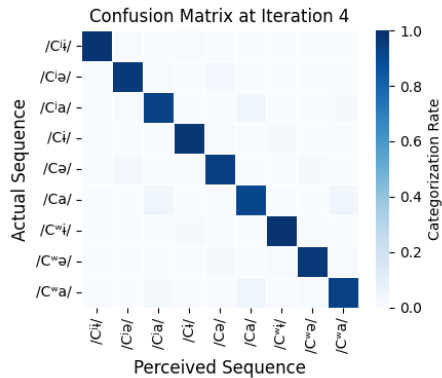
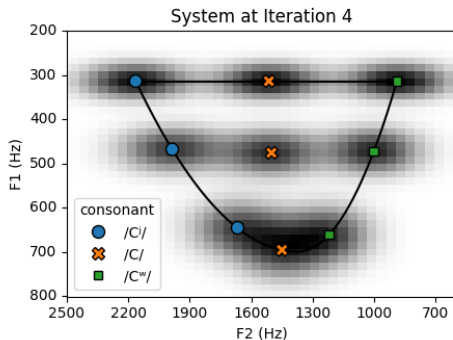
Case Study 2: Vertical Vowel Systems



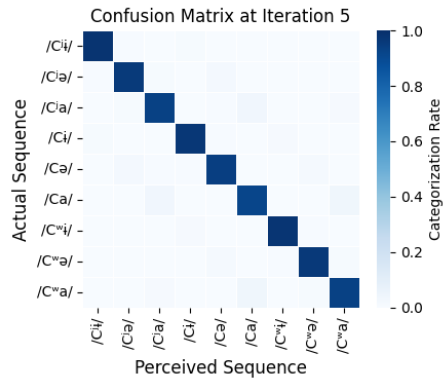
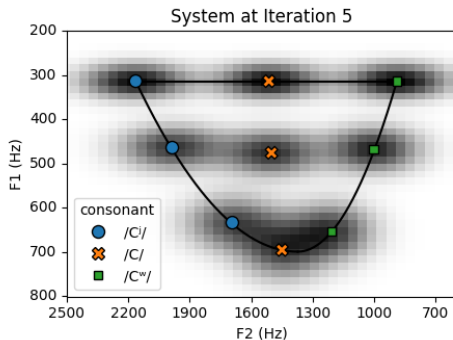
Case Study 2: Vertical Vowel Systems



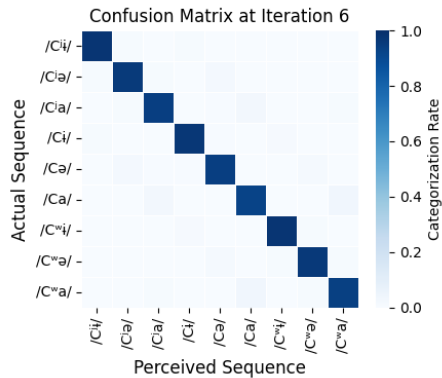
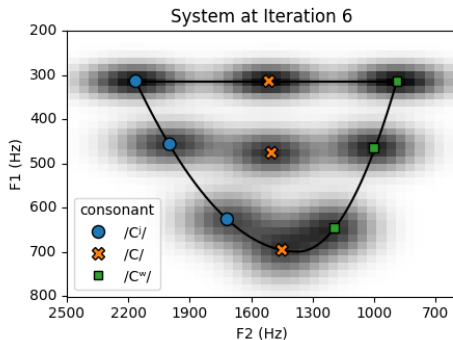
Case Study 2: Vertical Vowel Systems



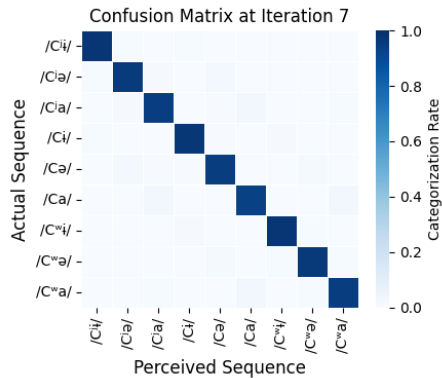
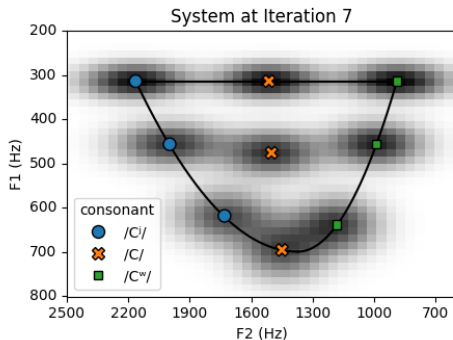
Case Study 2: Vertical Vowel Systems



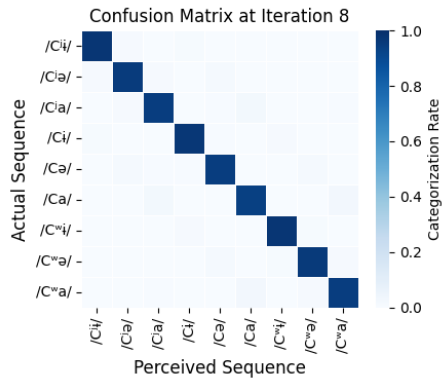
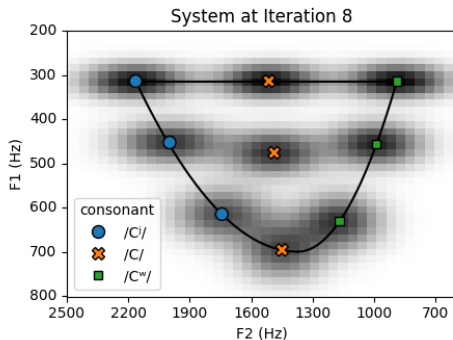
Case Study 2: Vertical Vowel Systems



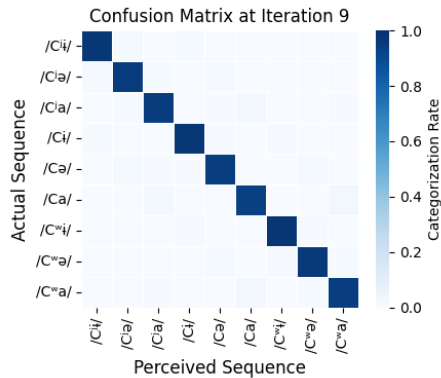
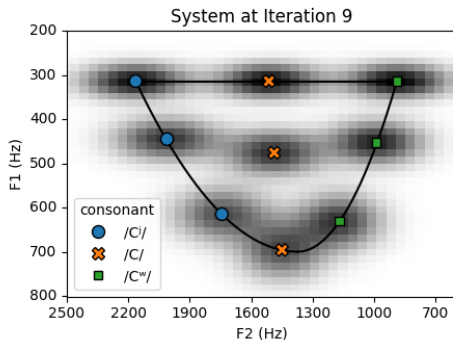
Case Study 2: Vertical Vowel Systems



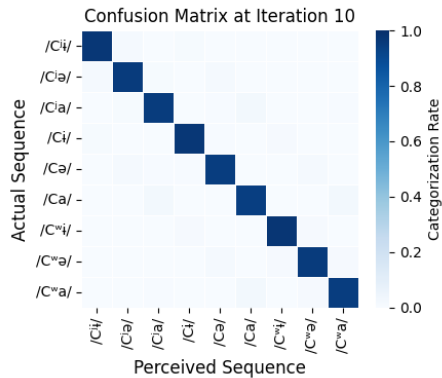
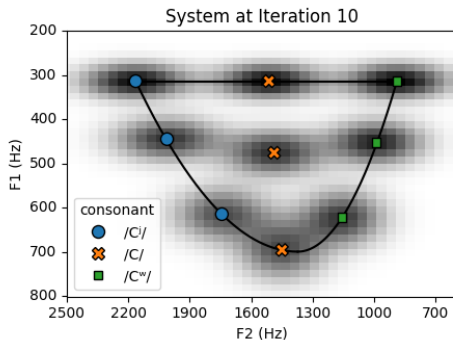
Case Study 2: Vertical Vowel Systems



Case Study 2: Vertical Vowel Systems

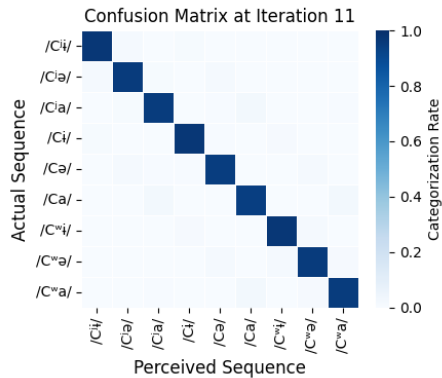
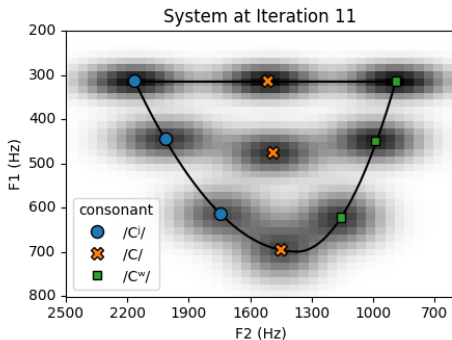


Case Study 2: Vertical Vowel Systems



Case Study 2: Vertical Vowel Systems

Vertical vowel systems stable as long as front & back allophones cue consonant contrast



Conclusions

Conclusions

Dispersion principles apply to not only segments but also sequences

Conclusions

Dispersion principles apply to not only segments but also sequences

- allows typological predictions regarding allophony

Conclusions

Dispersion principles apply to not only segments but also sequences

- allows typological predictions regarding allophony

Perceptually optimal sound structure can emerge from relatively superficial phonetic behaviors

Conclusions

Dispersion principles apply to not only segments but also sequences

- allows typological predictions regarding allophony

Perceptually optimal sound structure can emerge from relatively superficial phonetic behaviors

- Not all optimal structure needs to be encoded in innate grammar

Conclusions

Dispersion principles apply to not only segments but also sequences

- allows typological predictions regarding allophony

Perceptually optimal sound structure can emerge from relatively superficial phonetic behaviors

- Not all optimal structure needs to be encoded in innate grammar

Computational modeling of emergent structure can yield fine-grained predictions about:

Conclusions

Dispersion principles apply to not only segments but also sequences

- allows typological predictions regarding allophony

Perceptually optimal sound structure can emerge from relatively superficial phonetic behaviors

- Not all optimal structure needs to be encoded in innate grammar

Computational modeling of emergent structure can yield fine-grained predictions about:

- typological frequency, e.g. of vowel inventory types

Conclusions

Dispersion principles apply to not only segments but also sequences

- allows typological predictions regarding allophony

Perceptually optimal sound structure can emerge from relatively superficial phonetic behaviors

- Not all optimal structure needs to be encoded in innate grammar

Computational modeling of emergent structure can yield fine-grained predictions about:

- typological frequency, e.g. of vowel inventory types
- conditions required for given structure, e.g. vowel allophony primarily in small inventories

Conclusions

Dispersion principles apply to not only segments but also sequences

- allows typological predictions regarding allophony

Perceptually optimal sound structure can emerge from relatively superficial phonetic behaviors

- Not all optimal structure needs to be encoded in innate grammar

Computational modeling of emergent structure can yield fine-grained predictions about:

- typological frequency, e.g. of vowel inventory types
- conditions required for given structure, e.g. vowel allophony primarily in small inventories
- reasons for stability of typologically uncommon structure, e.g. vertical vowel systems

Acknowledgements

I would like to thank:

- Bruce Hayes, Megha Sundara, and the UCLA phonetics and phonology seminars for their thoughtful comments
- Research assistants Valerie So, Shawn Grant, Remi Akopians, Bill Tran, Makenna Kumlue, and Christopher Yang for their help creating the vowel acoustics data set used to fit the formant standard deviation model
- UCLA Division of Graduate Education's Graduate Research Mentorship program for funding

Conclusions

Dispersion principles apply to not only segments but also sequences

- allows typological predictions regarding allophony

Perceptually optimal sound structure can emerge from relatively superficial phonetic behaviors

- Not all optimal structure needs to be encoded in innate grammar

Computational modeling of emergent structure can yield fine-grained predictions about:

- typological frequency, e.g. of vowel inventory types
- conditions required for given structure, e.g. vowel allophony primarily in small inventories
- reasons for stability of typologically uncommon structure, e.g. vertical vowel systems

Conclusions

Dispersion principles apply to not only segments but also sequences

- allows typological predictions regarding allophony

Perceptually optimal sound structure can emerge from relatively superficial phonetic behaviors

- Not all optimal structure needs to be encoded in innate grammar

Computational modeling of emergent structure can yield fine-grained predictions about:

- typological frequency, e.g. of vowel inventory types
- conditions required for given structure, e.g. vowel allophony primarily in small inventories
- reasons for stability of typologically uncommon structure, e.g. vertical vowel systems

Thank You!

Implementing ICM over the Vowel Space

boundaries on the vowel space

We will consider vowel distributions with F1 and F2 means (μ_1, μ_2) in Hz limited by the following equations, adapted from Liljencrants & Lindblom (1972) to match the space reported by Becker-Kristal (2010) for /ieaou/:

$$314 \leq \mu_1 \leq 699 \quad (1)$$

$$\mu_2 \leq 1373 + (2164 - 1373) \sqrt{\frac{699 - \mu_1}{699 - 314}} \quad (2)$$

$$\mu_2 \geq 1373 + (889 - 1373) \sqrt{\frac{699 - \mu_1}{699 - 314}} \quad (3)$$

N.B. – Simulations here use Hz (not psychoacoustic scales used in past models) since phonetic overlap does not depend on choice of scale.

Implementing ICM over the Vowel Space

set of possible vowel distributions

Vowels are modeled as bivariate normal distributions with 0 covariance:

$$(F1, F2) \sim \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \right) \quad (4)$$

where:

- formant means (μ_1, μ_2) are limited by the previous equations, and
- standard deviations (σ_1, σ_2) are calculated as a function of formant means using a linear regression model fit to a crosslinguistic dataset.

Implementing ICM over the Vowel Space

formant standard deviation model

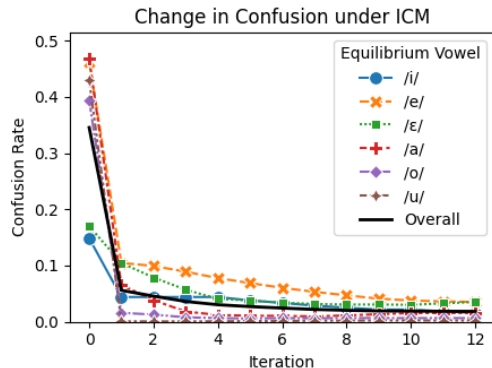
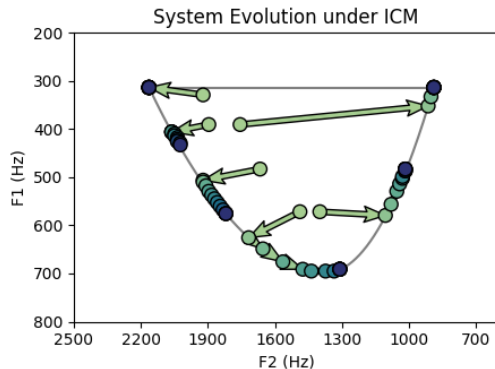
The standard deviation model was fit to z-scored data for 19 speakers of 21 languages from 10 studies^a with random effects for speaker and study:

$$\begin{aligned} z_1 &= \frac{\mu_1 - 498.7212}{163.5568} & z_2 &= \frac{\mu_2 - 1460.05}{543.4261} & (5) \\ \sigma_1 &= 34.9728 + 8.0738z_1 - 0.2633z_2 - 0.3906z_1z_2 \\ \sigma_2 &= \lambda(86.803 - 6.611z_1 + 6.741z_2 + 5.136z_1z_2) \end{aligned}$$

^aHoltse, 1972; Thongkum, 1986; McCord, 1989; Svantesson, 1995; Olson, 2001; Calamai et al., 2003; Calamai, 2004; Thornell and Nagano-Madsen, 2004; Chiang and Chiang, 2005; Mayr and Davies, 2009

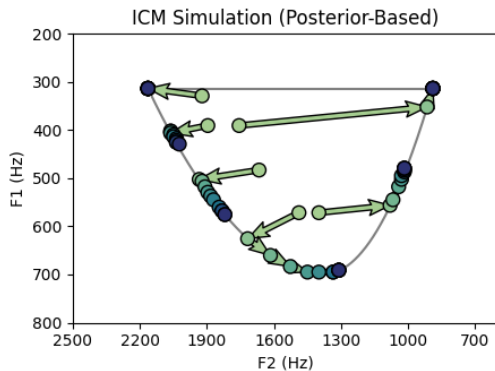
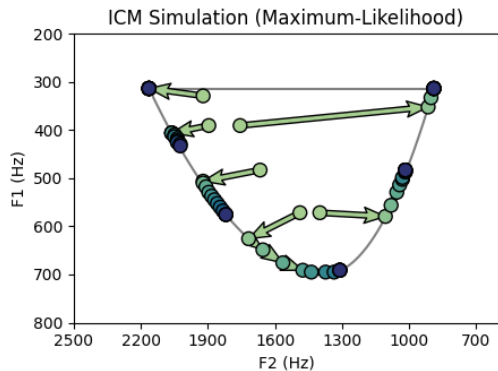
Following past dispersion models, this model also experiments with F2 scaling, implemented as a weight $1.0 \leq \lambda \leq 4.0$ on σ_2 .

Each ICM Iteration Reduces Overall Confusion



ICM with Maximum-Likelihood vs. Posterior-Tracking Behavior

More realistic probabilistic listener behavior does not significantly alter behavior of ICM



Consonant Likelihoods

Assume it is possible to discretize the phonetic space of consonant distributions into m regions

Given n consonant categories, let L be an $n \times m$ matrix encoding likelihood of a phonetic token in a region corresponding to a column given a consonant corresponding to a row

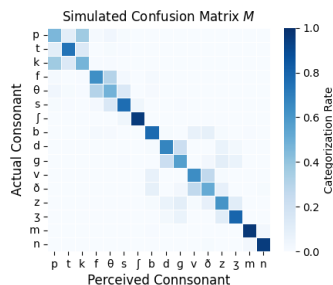
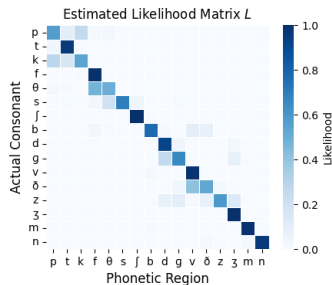
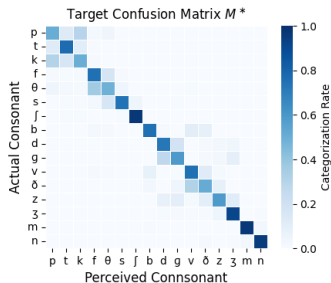
Assuming posterior-tracking behavior and a prior probability vector π over consonants, the expected $n \times n$ consonant confusion matrix M is defined:

$$M = L \operatorname{diag}(L^\top \pi)^{-1} L^\top \operatorname{diag}(\pi)$$

Given a confusion matrix M^* encoding our expectations about consonant confusability, we can work backwards from this equation using projected gradient descent to find a likelihood matrix L such that $M \approx M^*$

Consonant Likelihoods

Given a target confusion matrix M^* based on an experimental confusion matrix with 0 SNR (Miller and Nicely, 1955), we can find a likelihood matrix L yielding a simulation confusion matrix M that closely emulates the confusabilities encoding in M^* :



Works Cited I

- Choi, J.-D. (1992). *Phonetic underspecification and target-interpolation: an acoustic study of Marshallese vowel allophony*. University of California, Los Angeles.
- Clayards, M., Tanenhaus, M. K., Aslin, R. N., and Jacobs, R. A. (2008). Perception of speech reflects optimal use of probabilistic speech cues. *Cognition*, 108(3):804–809.
- de Boer, B. (1999). *Self organization in vowel systems*. PhD thesis, Ph. D. Dissertation, AI-Laboratory Vrije Universiteit Brussel.
- Feldman, N. H., Griffiths, T. L., and Morgan, J. L. (2009). The influence of categories on perception: explaining the perceptual magnet effect as optimal statistical inference. *Psychological review*, 116(4):752.
- Flemming, E. (1995). *Audio representations in phonology*. PhD thesis, Ph. D. Dissertation, University of California, Los Angeles.
- Flemming, E. (2004). Contrast and perceptual distinctiveness. *Phonetically based phonology*, 232:276.
- Gallagher, G. (2016). Vowel height allophony and dorsal place contrasts in cochabamba quechua. *Phonetica*, 73(2):101–119.
- Ganong, W. F. (1980). Phonetic categorization in auditory word perception. *Journal of experimental psychology: Human perception and performance*, 6(1):110.
- Kleinschmidt, D. F. (2019). Structure in talker variability: How much is there and how much can it help? *Language, cognition and neuroscience*, 34(1):43–68.
- Liljencrants, J. and Lindblom, B. (1972). Numerical simulation of vowel quality systems: The role of perceptual contrast. *Language*, pages 839–862.
- Lindblom, B. (1986). Phonetic universals in vowel systems. *Experimental phonology*, pages 13–44.
- Maddieson, I. (1984). *Pattern of Sounds*. Cambridge University Press, Cambridge, UK.
- Massaro, D. W. and Cohen, M. M. (1983). Phonological context in speech perception. *Perception & psychophysics*, 34(4):338–348.
- McCarthy, J. J. (1994). The phonetics and phonology of semitic pharyngeals. *Phonological structure and phonetic form*, 3:191–233.
- McGahay, J. (2025). Modeling vowel system typology using iterated confusion minimization. In *Proc. Interspeech 2025*, pages 2955–2959.
- McGahay, J. (under review). A theory of vowel system typology based on confusion minimization. *Language*. Manuscript under review.
- Miller, G. A. and Nicely, P. E. (1955). An analysis of perceptual confusions among some english consonants. *The Journal of the Acoustical Society of America*, 27(2):338–352.

Works Cited II

- Norris, D. and McQueen, J. M. (2008). Shortlist b: a bayesian model of continuous speech recognition. *Psychological review*, 115(2):357.
- Rischel, J. (1974). Topics in west greenlandic phonology. *Annual Report of the Institute of Phonetics University of Copenhagen*, 9:253–261.
- Schwartz, J.-L., Boë, L.-J., Vallée, N., and Abry, C. (1997a). Major trends in vowel system inventories. *Journal of phonetics*, 25(3):233–253.
- Schwartz, J.-L., Boë, L.-J., Vallée, N., and Abry, C. (1997b). The dispersion-focalization theory of vowel systems. *Journal of phonetics*, 25(3):255–286.
- Steffman, J. and Sundara, M. (2024). Disentangling the role of biphone probability from neighborhood density in the perception of nonwords. *Language and Speech*, 67(1):166–202.
- Steriade, D. (1993). “positional neutralization.” presentation at the 23rd meeting of the northeastern linguistic society. *University of Massachusetts, Amherst*.
- Steriade, D. (1997). Phonetics in phonology: the case of laryngeal neutralization. *Ms., UCLA*.
- Vaux, B. and Samuels, B. (2015). Explaining vowel systems: dispersion theory vs natural selection. *The Linguistic Review*, 32(3):573–599.